

MODULE II

FOURIER SERIES

Periodic function

A function $f(x)$ which satisfies the relation $f(x + T) = f(x)$ for all x and for a +ve constant T is a Periodic function. The smallest positive number T , for which this relation holds, is called the period of $f(x)$.

EXAMPLE: $\sin x, \cos x, \sec x$ and $\csc x$ are periodic functions with period 2π and those of $\tan x$ and $\cot x$ of period π . (See note 1)

Note: 1

$$\cos(x + 2\pi) = \cos x;$$

$$\sin(x + 2\pi) = \sin x; \sin(x + 4\pi) = \sin x; \sin(x + 6\pi) = \sin x$$

In general, $\sin(x + 2n\pi) = \sin x; n \in \mathbb{Z}$

$$\tan(x + \pi) = \tan x; \cot(x + \pi) = \cot x;$$

Dirichlet's Condition(Sufficient conditions for Fourier series)

A function $f(x)$ is expanded in a Fourier series in an interval $a \leq x \leq b$ if

- (i) $f(x)$ is well-defined and single valued except possibly at a finite number of points in (a, b) .
- (ii) $f(x)$ has a finite number of finite discontinuities and no infinite discontinuities in (a, b) and
- (iii) $f(x)$ has only a finite number of maxima and minima in (a, b) .

Fourier series

A periodic function $f(x)$ which satisfies certain conditions can be expanded as trigonometric series of the form

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \quad \text{--- (1)}$$

Where a_0, a_n and b_n are constants. The series on the right side of (1) is called the Fourier series.

Euler's formula for Fourier series

The Fourier series for the function $f(x)$ in the interval $c < x < c + 2l$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$$

$$\text{Where, } a_0 = \frac{1}{l} \int_c^{c+2l} f(x) dx$$

$$a_n = \frac{1}{l} \int_c^{c+2l} f(x) \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_c^{c+2l} f(x) \sin \frac{n\pi x}{l} dx$$

The above formula giving the values a_0, a_n and b_n are called the Euler formula or Fourier constant.

Convergence of Fourier series

(i) If $f(x)$ is continuous at $x = x_0$ then $f(x_0) = \lim_{x \rightarrow x_0} f(x)$.

(ii) If $f(x)$ has a discontinuity, the series converges to the average value of the right and left hand limits.

i.e., If $f(x)$ is discontinuous at $x = x_0$ then

$$\text{The sum of the Fourier series} = \frac{1}{2} [f(x_0 -) + f(x_0 +)]$$

Even and Odd Functions:

Analytically Even and Odd functions can be defined as follows.

Sl.	Function	Interval	Even	Odd
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No				
1.	$f(x)$	$(-l, l)$	$f(-x) = f(x)$	$f(-x) = -f(x)$
2.	$f(x) = \begin{cases} f_1(x), & (-l, 0) \\ f_2(x), & (0, l) \end{cases}$	$(-l, l)$	$f_1(-x) = f_2(x)$	$f_1(-x) = -f_2(x)$

3.	$f(x) = \begin{cases} f_1(x), & (-l, -\frac{l}{3}) \\ f_2(x), & (-\frac{l}{3}, \frac{l}{3}) \\ f_3(x), & (\frac{l}{3}, l) \end{cases}$	$(-l, l)$	$f_1(-x) = f_3(x)$ $f_2(-x) = f_2(x)$	$f_1(-x) = -f_3(x)$ $f_2(-x) = -f_2(x)$
4.	$f(x) = \begin{cases} f_1(x), & (-l, -\frac{l}{2}) \\ f_2(x), & (-\frac{l}{2}, 0) \\ f_3(x), & (0, \frac{l}{2}) \\ f_4(x), & (\frac{l}{2}, l) \end{cases}$	$(-l, l)$	$f_1(-x) = f_4(x)$ $f_2(-x) = f_3(x)$	$f_1(-x) = -f_4(x)$ $f_2(-x) = -f_3(x)$

Examples of Even function: $f(x) = x^2$

$$f(-x) = (-x)^2 = x^2 = f(x)$$

$\therefore f(x) = x^2$ is an even function.

Also $f(x) = \cos x, |x|, |\cos x|, |\sin x|, x^2, x^4, \dots$, are even functions.

The graphs of an even function always symmetrically with respect to y-axis.

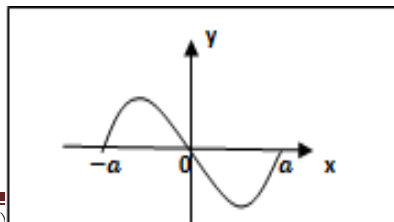
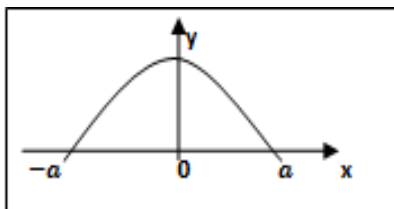
Advantage : $\int_{-a}^a f(x)dx = 2 \int_0^a f(x) dx$ if $f(x)$ is an even function

Examples of Odd function: $f(x) = \sin x$

$$f(-x) = \sin(-x) = -\sin x = -f(x)$$

$\therefore f(x) = \sin x$ is an odd function.

Also $f(x) = x, \tan x, x^3$ are odd functions.



The graph of an odd function always symmetric in opposite quadrants.

Advantage: $\int_{-a}^a f(x)dx = 0$ if $f(x)$ is an odd function

Note : Even function $\times$ Even function = Even function

Even function $\times$ Odd function = Odd function

Odd function $\times$ Even function = Odd function

Odd function $\times$ Odd function = Even function

Bernoulli's Generalized Rule

If u is the algebraic function and v is the trigonometric function, then

$$\int uv = uv_1 - u'v_2 + u''v_3 - \dots$$

Where **dashes** denotes differentiation & **suffixes** denotes integration.

Example : $\int x^2 \cos x dx = [(x^2)(\sin x) - (2x)(-\cos x) + (2)(-\sin x)]$
 $= x^2 \sin x + 2x \cos x - 2 \sin x$

Useful Results

When 'n' is an integer 1. $\sin n\pi = 0$ 2. $\cos n\pi = (-1)^n$

Change of Interval

Fourier series in $(0, 2l)$

The Fourier series of $f(x)$ in the interval $0 \leq x \leq 2l$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l} \quad \text{--- (A)}$$

where $a_0 = \frac{1}{l} \int_0^{2l} f(x) dx$,

$$a_n = \frac{1}{l} \int_0^{2l} f(x) \cos \frac{n\pi x}{l} dx \text{ \& }$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin \frac{n\pi x}{l} dx$$

Fourier series in $(-l, l)$

The Fourier series of $f(x)$ in the interval $-l \leq x \leq l$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l} \quad \text{--- (B)}$$

where, $a_0 = \frac{1}{l} \int_{-l}^l f(x) dx$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx \quad \& \quad b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx$$

Case: 1 Fourier series of Even Function in $(-l, l)$

If $f(x)$ is an even function of x in $(-l, l)$, then the Fourier expansion of $f(x)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l}$$

where, $a_0 = \frac{2}{l} \int_0^l f(x) dx$ & $a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx$

Case: 2 Fourier series of Odd Function in $(-l, l)$

If $f(x)$ is an odd function of x in $(-l, l)$, then the Fourier expansion of $f(x)$ is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$$

where, $b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$

ILLUSTRATIVE EXAMPLES UNDER $(0, 2l)$

Example:1 Find the Fourier series of $f(x) = x(2l - x)$ in $(0, 2l)$. Deduce the sum of $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots$

Solution: Given $f(x) = x(2l - x) = 2lx - x^2$

The Fourier series of $f(x)$ in the interval $0 \leq x \leq 2l$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$$

where, $a_0 = \frac{1}{l} \int_0^{2l} f(x) dx$, $a_n = \frac{1}{l} \int_0^{2l} f(x) \cos \frac{n\pi x}{l} dx$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin \frac{n\pi x}{l} dx$$

To find a_0

$$\begin{aligned} a_0 &= \frac{1}{l} \int_0^{2l} f(x) dx \\ &= \frac{1}{l} \int_0^{2l} (2lx - x^2) dx = \frac{1}{l} \left[\frac{2lx^2}{2} - \frac{x^3}{3} \right]_0^{2l} = \frac{1}{l} \left[4l^3 - \frac{8l^3}{3} \right] = \frac{4l^2}{3} \end{aligned}$$

To find a_n :

$$\begin{aligned} a_n &= \frac{1}{l} \int_0^{2l} f(x) \cos \frac{n\pi x}{l} dx = \frac{1}{l} \int_0^{2l} (2lx - x^2) \cos \frac{n\pi x}{l} dx \\ &= \frac{1}{l} \left\{ (2lx - x^2) \left(\frac{\sin \frac{n\pi x}{l}}{\frac{n\pi}{l}} \right) - (2l - 2x) \left(\frac{-\cos \frac{n\pi x}{l}}{\frac{n^2 \pi^2}{l^2}} \right) + (-2) \left(\frac{-\sin \frac{n\pi x}{l}}{\frac{n^3 \pi^3}{l^3}} \right) \right\}_0^{2l} \\ &= \frac{1}{l} \left\{ \left[(2lx - x^2) \left(\frac{l}{n\pi} \right) \left(\sin \frac{n\pi x}{l} \right) + (2l - 2x) \left(\frac{l^2}{n^2 \pi^2} \right) \cos \frac{n\pi x}{l} \right. \right. \\ &\quad \left. \left. + 2 \left(\frac{l^3}{n^3 \pi^3} \right) \left(\sin \frac{n\pi x}{l} \right) \right]_0^{2l} \right\} \end{aligned}$$

$$= \frac{1}{l} \left\{ \left[0 + (-2l) \frac{l^2}{n^2 \pi^2} + 0 \right] - \left[0 + (2l) \frac{l^2}{n^2 \pi^2} + 0 \right] \right\} \begin{cases} \because \cos 2n\pi = 1 \\ \& \sin n\pi = 0 \end{cases}$$

$$= \frac{1}{l} \left[-\frac{4l^3}{n^2 \pi^2} \right]$$

$$a_n = -\frac{4l^2}{n^2 \pi^2}$$

To find b_n :

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin \frac{n\pi x}{l} dx$$

$$= \frac{1}{l} \int_0^{2l} (2lx - x^2) \sin \frac{n\pi x}{l} dx$$

$$= \frac{1}{l} \left\{ (2lx - x^2) \left(\frac{-\cos \frac{n\pi x}{l}}{\frac{n\pi}{l}} \right) - (2l - 2x) \left(\frac{-\sin \frac{n\pi x}{l}}{\frac{n^2 \pi^2}{l^2}} \right) + (-2) \left(\frac{\cos \frac{n\pi x}{l}}{\frac{n^3 \pi^3}{l^3}} \right) \right\}_0^{2l}$$

$$= \frac{1}{l} \left\{ \left[-(2lx - x^2) \left(\frac{l}{n\pi} \right) \left(\cos \frac{n\pi x}{l} \right) + (2l - 2x) \left(\frac{l^2}{n^2 \pi^2} \right) \sin \frac{n\pi x}{l} - 2 \left(\frac{l^3}{n^3 \pi^3} \right) \left(\cos \frac{n\pi x}{l} \right) \right]_0^{2l} \right\}$$

$$= \frac{1}{l} \left\{ \left[0 + 0 - \frac{2l^3}{n^3 \pi^3} \right] - \left[0 + 0 - \frac{2l^3}{n^3 \pi^3} \right] \right\}$$

$$b_n = 0$$

Hence, the required Fourier series is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$$

$$= \frac{2l^2}{3} - \frac{4l^2}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \cos \frac{n\pi x}{l} \quad \text{--- (1)}$$

Put $x = l$ in (1), we get

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$$f(x) = x(2l - x) \text{ in } (0, 2l)$$

$$\text{Left hand limit } f(0) = 0$$

$$f(0) = f(2l) \text{ i.e., equal}$$

$$f(l) = \frac{2l^2}{3} - \frac{4l^2}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \cos n\pi$$

At $x = l$, the function $f(x)$ is continuous

$$l(2l - l) = \frac{2l^2}{3} - \frac{4l^2}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$

$$l^2 - \frac{2l^2}{3} = -\frac{4l^2}{\pi^2} \left(-\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \dots \right)$$

$$\left(\frac{l^2}{3} \right) \left(\frac{\pi^2}{4l^2} \right) = - \left(-\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \dots \right)$$

$$\therefore \frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots$$

Example: 2 Find the Fourier series of $f(x)$ with period $2l$, defined by

$$f(x) = \begin{cases} l - x, & 0 < x < l \\ 0, & l < x < 2l \end{cases} \text{ Hence deduce that } \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}.$$

$$\text{Solution: Given } f(x) = \begin{cases} l - x, & 0 < x < l \\ 0, & l < x < 2l \end{cases}$$

The Fourier series of $f(x)$ in $0 \leq x \leq 2l$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l} \quad \text{--- (1)}$$

$$\text{where, } a_0 = \frac{1}{l} \int_0^{2l} f(x) dx; \quad a_n = \frac{1}{l} \int_0^{2l} f(x) \cos \frac{n\pi x}{l} dx \&$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin \frac{n\pi x}{l} dx$$

To find a_0 :

$$a_0 = \frac{1}{l} \int_0^{2l} f(x) dx$$

$$\begin{aligned}
&= \frac{1}{l} \left\{ \int_0^l f(x) dx + \int_l^{2l} f(x) dx \right\} \\
&= \frac{1}{l} \left\{ \int_0^l (l-x) dx + \int_l^{2l} (0) dx \right\} \\
&= \frac{1}{l} \left\{ lx - \frac{x^2}{2} \right\}_0^l = \frac{1}{l} \left[l^2 - \frac{l^2}{2} \right] = \frac{1}{l} \left[\frac{l^2}{2} \right]
\end{aligned}$$

$$a_0 = \frac{l}{2}$$

To find a_n :

$$\begin{aligned}
a_n &= \frac{1}{l} \int_0^{2l} f(x) \cos \frac{n\pi x}{l} dx \\
&= \frac{1}{l} \int_0^l (l-x) \cos \frac{n\pi x}{l} dx \\
&= \frac{1}{l} \left\{ (l-x) \left(\frac{\sin \frac{n\pi x}{l}}{\frac{n\pi}{l}} \right) - (-1) \left(\frac{-\cos \frac{n\pi x}{l}}{\frac{n^2\pi^2}{l^2}} \right) \right\}_0^l \\
&= \frac{1}{l} \left\{ \left[(l-x) \left(\frac{l}{n\pi} \right) \left(\sin \frac{n\pi x}{l} \right) - \left(\frac{l^2}{n^2\pi^2} \right) \cos \frac{n\pi x}{l} \right]_0^l \right\} \\
&= \frac{1}{l} \left\{ \left[0 - \frac{l^2}{n^2\pi^2} (-1)^n \right] - \left[0 - \frac{l^2}{n^2\pi^2} \right] \right\} \\
&= \frac{l}{n^2\pi^2} \{ 1 - (-1)^n \}
\end{aligned}$$

$$a_n = \begin{cases} \frac{2l}{n^2\pi^2}, & \text{when } n \text{ is odd} \\ 0, & \text{when } n \text{ is even} \end{cases}$$

To find b_n :

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin \frac{n\pi x}{l} dx$$

$$\because \cos n\pi = (-1)^n \\ \& \sin n\pi = 0$$

$$\begin{aligned}
&= \frac{1}{l} \int_0^l (l-x) \sin \frac{n\pi x}{l} dx \\
&= \frac{1}{l} \left\{ (l-x) \left(\frac{-\cos \frac{n\pi x}{l}}{\frac{n\pi}{l}} \right) - (-1) \left(\frac{-\sin \frac{n\pi x}{l}}{\frac{n^2\pi^2}{l^2}} \right) \right\}_0^l \\
&= \frac{1}{l} \left\{ [0 - 0] - \left[-\frac{l^2}{n\pi} - 0 \right] \right\} \\
&= \frac{1}{l} \left\{ -(l-x) \left(\frac{l}{n\pi} \right) \left(\cos \frac{n\pi x}{l} \right) - \left(\frac{l^2}{n^2\pi^2} \right) \sin \frac{n\pi x}{l} \right\}_0^l
\end{aligned}$$

$$b_n = \frac{l}{n\pi}$$

Hence, the required Fourier series is

$$\begin{aligned}
f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l} \\
&= \frac{l}{4} + \frac{2l}{\pi^2} \sum_{n=1,3,\dots}^{\infty} \frac{1}{n^2} \cos \frac{n\pi x}{l} + \frac{l}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x}{l} \quad \text{--- (2)}
\end{aligned}$$

Put $x = 0$ in (2), we get

$$f(0) = \frac{l}{4} + \frac{2l}{\pi^2} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2}$$

At $x = 0$, the function $f(x)$ is **discontinuous**

$$\left(\frac{\text{left hand limit} + \text{right hand limit}}{2} \right) = \frac{l}{4} + \frac{2l}{\pi^2} \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right)$$

$$\frac{l+0}{2} = \frac{l}{4} + \frac{2l}{\pi^2} \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right)$$

$$\frac{l}{2} - \frac{l}{4} = \frac{2l}{\pi^2} \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right)$$

$$\left(\frac{l}{4} \right) \left(\frac{\pi^2}{2l} \right) = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

$$\begin{aligned}
f(x) &= \begin{cases} l-x, & 0 < x < l \\ 0, & l < x < 2l \end{cases} \\
\text{Left hand limit } f(0) &= l-0 = l \\
\text{Right hand limit } f(2l) &= 0 \\
f(0) &\neq f(2l) \text{ i.e., unequal} \\
\therefore f(x) &\text{ is discontinuous}
\end{aligned}$$

$$\therefore \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$$

Example: 3 Obtain Fourier series for the function $f(x) = \begin{cases} x, & 0 < x < 1 \\ (2-x), & 1 < x < 2 \end{cases}$

Hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$

Solution: Here $2l = 2 \Rightarrow l = 1$

The Fourier series of $f(x)$ in $0 \leq x \leq 2$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n\pi x + \sum_{n=1}^{\infty} b_n \sin n\pi x \quad \text{--- (1)}$$

Where $a_0 = \int_0^2 f(x) dx$; $a_n = \int_0^2 f(x) \cos n\pi x dx$ &

$$b_n = \int_0^2 f(x) \sin n\pi x dx \quad \because l = 1$$

To find a_0 :

$$\begin{aligned} a_0 &= \int_0^2 f(x) dx \\ &= \int_0^1 f(x) dx + \int_1^2 f(x) dx \\ &= \int_0^1 x dx + \int_1^2 (2-x) dx \\ &= \left[\frac{x^2}{2} \right]_0^1 + \left[2x - \frac{x^2}{2} \right]_1^2 = \left[\frac{1}{2} - 0 \right] + \left[2 - \frac{3}{2} \right] \\ &= \frac{1}{2} + \frac{1}{2} \end{aligned}$$

$$a_0 = 1$$

To find a_n :

$$\begin{aligned} a_n &= \int_0^2 f(x) \cos n\pi x dx \\ &= \int_0^1 x \cos n\pi x dx + \int_1^2 (2-x) \cos n\pi x dx \\ &= \left\{ \left[(x) \left(\frac{\sin n\pi x}{n\pi} \right) - (1) \left(\frac{-\cos n\pi x}{n^2 \pi^2} \right) \right]_0^1 \right. \\ &\quad \left. + \left[(2-x) \left(\frac{\sin n\pi x}{n\pi} \right) - (-1) \left(\frac{-\cos n\pi x}{n^2 \pi^2} \right) \right]_1^2 \right\} \\ &= \left\{ \left[(x) \left(\frac{\sin n\pi x}{n\pi} \right) + \left(\frac{\cos n\pi x}{n^2 \pi^2} \right) \right]_0^1 + \left[(2-x) \left(\frac{\sin n\pi x}{n\pi} \right) - \left(\frac{\cos n\pi x}{n^2 \pi^2} \right) \right]_1^2 \right\} \\ &= \left\{ \left[\left(0 + \frac{(-1)^n}{n^2 \pi^2} \right) - \left(0 + \frac{1}{n^2 \pi^2} \right) \right] + \left[\left(0 - \frac{1}{n^2 \pi^2} \right) - \left(0 - \frac{(-1)^n}{n^2 \pi^2} \right) \right] \right\} \\ &= \frac{2(-1)^n}{n^2 \pi^2} - \frac{2}{n^2 \pi^2} \\ &= \frac{2}{n^2 \pi^2} \{ (-1)^n - 1 \} \end{aligned}$$

$$a_n = \begin{cases} \frac{-4}{n^2 \pi^2}, & \text{when } n \text{ is odd} \\ 0, & \text{when } n \text{ is even} \end{cases}$$

To find b_n

$$\begin{aligned} b_n &= \int_0^2 f(x) \sin n\pi x dx \\ &= \int_0^1 f(x) \sin n\pi x dx + \int_1^2 f(x) \sin n\pi x dx \end{aligned}$$

$$= \int_0^1 x \sin n\pi x dx + \int_1^2 (2-x) \sin n\pi x dx$$

$$= \left\{ \left[(x) \left(\frac{-\cos n\pi x}{n\pi} \right) - (1) \left(\frac{-\sin n\pi x}{n^2 \pi^2} \right) \right]_0^1 + \left[(2-x) \left(\frac{-\cos n\pi x}{n\pi} \right) - (-1) \left(\frac{-\sin n\pi x}{n^2 \pi^2} \right) \right]_1^2 \right\}$$

$$= \left\{ \left[- (x) \left(\frac{\cos n\pi x}{n\pi} \right) + \left(\frac{\sin n\pi x}{n^2 \pi^2} \right) \right]_0^1 + \left[-(2-x) \left(\frac{\cos n\pi x}{n\pi} \right) - \left(\frac{\sin n\pi x}{n^2 \pi^2} \right) \right]_1^2 \right\}$$

$$= \left\{ \left[\left(-\frac{(-1)^n}{n\pi} + 0 \right) - (0-0) \right] + \left[(0-0) - \left(-\frac{(-1)^n}{n\pi} - 0 \right) \right] \right\}$$

$$= -\frac{(-1)^n}{n\pi} + \frac{(-1)^n}{n\pi}$$

$$b_n = 0$$

Hence, the required Fourier series (1) becomes

$$f(x) = \frac{1}{2} + \left(-\frac{4}{\pi^2} \right) \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2} \cos n\pi x$$

Put $x = 0$ in the above, we get

$$f(0) = \frac{1}{2} + \left(\frac{-4}{\pi^2} \right) \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2} \quad \text{--- (2)}$$

At $x = 0$, the function $f(x)$ is continuous

$$\therefore (2) \text{ becomes, } 0 = \frac{1}{2} + \left(\frac{-4}{\pi^2} \right) \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2}$$

$$\left(-\frac{1}{2} \right) \left(\frac{\pi^2}{-4} \right) = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

$$\therefore \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$$

Example : 4 Obtain the Fourier series of periodicity 3 for $f(x) = 2x - x^2$, $0 < x < 3$

Solution: Given $f(x) = 2x - x^2$, $0 < x < 3$

$$\text{Here } 2l = 3 \Rightarrow l = \frac{3}{2}$$

The Fourier series of $f(x)$ in $0 \leq x \leq 3$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{2n\pi x}{3} + \sum_{n=1}^{\infty} b_n \sin \frac{2n\pi x}{3} \quad \text{--- (1)}$$

Where,

$$a_0 = \frac{2}{3} \int_0^3 f(x) dx; \quad a_n = \frac{2}{3} \int_0^3 f(x) \cos \frac{2n\pi x}{3} dx \text{ \& } b_n = \frac{2}{3} \int_0^3 f(x) \sin \frac{2n\pi x}{3} dx$$

To find a_0 :

$$a_0 = \frac{2}{3} \int_0^3 f(x) dx$$

$$= \frac{2}{3} \int_0^3 (2x - x^2) dx = \frac{2}{3} \left[2 \left(\frac{x^2}{2} \right) - \frac{x^3}{3} \right]_0^3 = \frac{2}{3} \left[9 - \frac{27}{3} \right] = \frac{2}{3} [9 - 9]$$

$$a_0 = 0$$

To find a_n :

$$a_n = \frac{2}{3} \int_0^3 f(x) \cos \frac{2n\pi x}{3} dx = \frac{2}{3} \int_0^3 (2x - x^2) \cos \frac{2n\pi x}{3} dx$$

$$= \frac{2}{3} \left\{ (2x - x^2) \left(\frac{\sin \frac{2n\pi x}{3}}{\frac{2n\pi}{3}} \right) - (2 - 2x) \left(\frac{-\cos \frac{2n\pi x}{3}}{\left(\frac{2n\pi}{3} \right)^2} \right) + (-2) \left(\frac{-\sin \frac{2n\pi x}{3}}{\left(\frac{2n\pi}{3} \right)^3} \right) \right\}_0^3$$

$$= \frac{2}{3} \left\{ (2x - x^2) \left(\frac{3}{2n\pi} \right) \left(\sin \frac{2n\pi x}{3} \right) + (2 - 2x) \left(\frac{3}{2n\pi} \right)^2 \left(\cos \frac{2n\pi x}{3} \right) + (2) \left(\frac{3}{2n\pi} \right)^3 \left(\sin \frac{2n\pi x}{3} \right) \right\}_0^3$$

$$= \frac{2}{3} \left\{ \left(0 - 4 \left(\frac{3}{2n\pi} \right)^2 + 0 \right) - \left(0 + 2 \left(\frac{3}{2n\pi} \right)^2 + 0 \right) \right\}$$

$$= \frac{2}{3} \left\{ -6 \left(\frac{3}{2n\pi} \right)^2 \right\} = \frac{2}{3} \left\{ -6 \left(\frac{9}{4n^2\pi^2} \right) \right\} = -4 \left(\frac{9}{4n^2\pi^2} \right) a_n = -\frac{9}{n^2\pi^2}$$

To find b_n :

$$\begin{aligned} b_n &= \frac{2}{3} \int_0^3 f(x) \sin \frac{2n\pi x}{3} dx \\ &= \frac{2}{3} \int_0^3 (2x - x^2) \sin \frac{2n\pi x}{3} dx \\ &= \frac{2}{3} \left\{ (2x - x^2) \left(\frac{-\cos \frac{2n\pi x}{3}}{\frac{2n\pi}{3}} \right) - (2 - 2x) \left(\frac{-\sin \frac{2n\pi x}{3}}{\left(\frac{2n\pi}{3} \right)^2} \right) + (-2) \left(\frac{\cos \frac{2n\pi x}{3}}{\left(\frac{2n\pi}{3} \right)^3} \right) \right\}_0^3 \\ &= \frac{2}{3} \left\{ -(2x - x^2) \left(\frac{3}{2n\pi} \right) \left(\cos \frac{2n\pi x}{3} \right) + (2 - 2x) \left(\frac{3}{2n\pi} \right)^2 \left(\sin \frac{2n\pi x}{3} \right) \right. \\ &\quad \left. - (2) \left(\frac{3}{2n\pi} \right)^3 \left(\cos \frac{2n\pi x}{3} \right) \right\}_0^3 \\ &= \frac{2}{3} \left\{ \left(3 \left(\frac{3}{2n\pi} \right) + 0 - 2 \left(\frac{3}{2n\pi} \right)^3 \right) - \left(0 + 0 - 2 \left(\frac{3}{2n\pi} \right)^3 \right) \right\} \\ &= \frac{2}{3} \left\{ \frac{9}{2n\pi} - 2 \left(\frac{3}{2n\pi} \right)^3 + 2 \left(\frac{3}{2n\pi} \right)^3 \right\} = \frac{2}{3} \left\{ \frac{9}{2n\pi} \right\} \\ b_n &= \frac{3}{n\pi} \end{aligned}$$

$\therefore$ (1) becomes

$$f(x) = -\frac{9}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \cos \frac{2n\pi x}{3} + \frac{3}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{2n\pi x}{3}$$

Practice problems

1. Find the Fourier series of function $f(x) = \begin{cases} kx, & 0 < x < l \\ 0, & l < x < 2l \end{cases}$ k being a constant.

$$\text{Ans : } a_0 = \frac{kl}{2}; a_n = \begin{cases} \frac{-2kl}{n^2\pi^2} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}; b_n = \frac{-kl(-1)^n}{n\pi}$$

2. Find the Fourier expansion of period $2l$ for the function $f(x) = (l - x)^2$ in the range $(0, 2l)$. Deduce the sum of the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$.

$$\text{Ans : } a_0 = \frac{2}{3}l^2; a_n = \frac{4l^2}{n^2\pi^2}; b_n = 0$$

3. Obtain the Fourier series of the function $f(x) = 2x - x^2, 0 \leq x \leq 2$

Hint: here $2l = 2 \Rightarrow l = 1$

$$\text{Ans : } a_0 = \frac{4}{3}; a_n = \frac{-4}{n^2\pi^2}; b_n = 0$$

4. Find the Fourier series for the function $f(x) = \begin{cases} x, & 0 < x < 1 \\ 1 - x, & 1 < x < 2 \end{cases}$ and deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$

$$\text{Ans : } a_0 = 0; a_n = \begin{cases} \frac{-4}{n^2\pi^2} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}; b_n = \begin{cases} \frac{2}{n\pi} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

ILLUSTRATIVE EXAMPLES UNDER $(-l, l)$

Example: 1 Obtain the Fourier series for the function

$$f(x) = \begin{cases} 1 + \frac{2x}{l}, & -l \leq x \leq 0 \\ 1 - \frac{2x}{l}, & 0 \leq x \leq l \end{cases}. \text{ Hence deduce that } \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$$

$$\text{Solution: Given } f(x) = \begin{cases} 1 + \frac{2x}{l} & \text{in } -l \leq x \leq 0 \\ 1 - \frac{2x}{l} & \text{in } 0 \leq x \leq l \end{cases}$$

$$\text{Let } f_1(x) = 1 + \frac{2x}{l}; f_2(x) = 1 - \frac{2x}{l}$$

$$f_1(-x) = 1 + \frac{2(-x)}{l} = 1 - \frac{2x}{l} = f_2(x)$$

$\therefore f(x)$ is an even function. The Fourier series of $f(x)$ contains cosine terms only, and hence $b_n = 0$.

The Fourier series of $f(x)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} \quad \text{--- (1)}$$

where $a_0 = \frac{2}{l} \int_0^l f(x) dx$ & $a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx$

To find a_0 :

$$\begin{aligned} a_0 &= \frac{2}{l} \int_0^l f(x) dx \\ &= \frac{2}{l} \int_0^l \left(1 - \frac{2x}{l}\right) dx \\ &= \frac{2}{l} \left\{ x - \frac{2}{l} \left(\frac{x^2}{2} \right) \right\}_0^l = \frac{2}{l} \left\{ \left(l - \frac{l^2}{l} \right) - (0 - 0) \right\} \end{aligned}$$

$$a_0 = 0$$

To find a_n

$$\begin{aligned} a_n &= \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx = \frac{2}{l} \int_0^l \left(1 - \frac{2x}{l}\right) \cos \frac{n\pi x}{l} dx \\ &= \frac{2}{l} \left\{ \left(1 - \frac{2x}{l}\right) \left(\frac{\sin \frac{n\pi x}{l}}{\frac{n\pi}{l}} \right) - \left(-\frac{2}{l}\right) \left(\frac{-\cos \frac{n\pi x}{l}}{\left(\frac{n\pi}{l}\right)^2} \right) \right\}_0^l \\ &= \frac{2}{l} \left\{ \left(1 - \frac{2x}{l}\right) \left(\frac{l}{n\pi} \right) \left(\sin \frac{n\pi x}{l} \right) - \left(\frac{2}{l} \right) \left(\frac{l}{n\pi} \right)^2 \left(\cos \frac{n\pi x}{l} \right) \right\}_0^l \\ &= \frac{2}{l} \left\{ \left(0 - \frac{2l^2}{ln^2\pi^2} (-1)^n \right) - \left(0 - \frac{2l^2}{ln^2\pi^2} \right) \right\} \\ &= \frac{2}{l} \left\{ -\frac{2l}{n^2\pi^2} (-1)^n + \frac{2l}{n^2\pi^2} \right\} = \frac{2}{l} \cdot \frac{2l}{n^2\pi^2} [1 - (-1)^n] \end{aligned}$$

$$= \frac{4}{n^2\pi^2} [1 - (-1)^n]$$

$$a_n = \begin{cases} \frac{8}{n^2\pi^2} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

$$\begin{aligned} \therefore (1) \text{ becomes, } f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} \\ &= \frac{8}{\pi^2} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2} \cos \frac{n\pi x}{l} \end{aligned} \quad \text{--- (2)}$$

At $x = 0$, $f(x)$ is continuous then (2) becomes

$$\begin{aligned} f(0) &= \frac{8}{\pi^2} \left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \dots \dots \infty \right] \\ 1 &= \frac{8}{\pi^2} \left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \dots \dots \infty \right] \\ \therefore \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \dots \dots \infty &= \frac{\pi^2}{8} \end{aligned}$$

$$f(x) = \begin{cases} 1 + \frac{2x}{l} & -l \leq x \leq 0 \\ 1 - \frac{2x}{l} & 0 \leq x \leq l \end{cases}$$

Right hand limit $f(0-) = 1 + 0 = 1$
 Left hand limit $f(0+) = 1 - 0 = 1$
 $f(0-) = f(0+)$ i.e., equal
 $\therefore f(x)$ is continuous

Example: 2 Find the Fourier series to represent $f(x) = x^2 - 2$, $-2 < x < 2$.

Hence deduce that $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots$

Solution : Given $f(x) = x^2 - 2$, $-2 < x < 2$

$$f(-x) = (-x)^2 - 2 = x^2 - 2 = f(x)$$

$\therefore f(x)$ is an even function. The Fourier series of $f(x)$ contains cosine terms only, and hence $b_n = 0$. Here $2l = 4 \Rightarrow l = 2$

The Fourier series of $f(x)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{2} \quad \text{--- (1)}$$

$$\text{where } a_0 = \int_0^2 f(x) dx \text{ \& } a_n = \int_0^2 f(x) \cos \frac{n\pi x}{2} dx \quad \because l = 2$$

To find a_0 :

$$\begin{aligned} a_0 &= \int_0^2 f(x) dx \\ &= \int_0^2 (x^2 - 2) dx = \left\{ \frac{x^3}{3} - 2x \right\}_0^2 = \frac{8}{3} - 4 \end{aligned}$$

$$a_0 = -\frac{4}{3}$$

To find a_n :

$$\begin{aligned} a_n &= \int_0^2 f(x) \cos \frac{n\pi x}{2} dx \\ &= \int_0^2 (x^2 - 2) \cos \frac{n\pi x}{2} dx \\ &= \left\{ (x^2 - 2) \left(\frac{\sin \frac{n\pi x}{2}}{\frac{n\pi}{2}} \right) - (2x) \left(\frac{-\cos \frac{n\pi x}{2}}{\left(\frac{n\pi}{2} \right)^2} \right) + (2) \left(\frac{-\sin \frac{n\pi x}{2}}{\left(\frac{n\pi}{2} \right)^3} \right) \right\}_0^2 \\ &= \left\{ (x^2 - 2) \left(\frac{2}{n\pi} \right) \left(\sin \frac{n\pi x}{2} \right) + (2x) \left(\frac{2}{n\pi} \right)^2 \left(\cos \frac{n\pi x}{2} \right) - (2) \left(\frac{2}{n\pi} \right)^3 \left(\sin \frac{n\pi x}{2} \right) \right\}_0^2 \\ &= \left\{ \left(0 + \frac{16}{n^2 \pi^2} (-1)^n - 0 \right) - (0 + 0 - 0) \right\} \{ \because \cos n\pi = (-1)^n \} \end{aligned}$$

$$a_n = \frac{16}{n^2 \pi^2} (-1)^n$$

$\therefore$ (1) becomes

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{2} \{ \because l = 2 \}$$

$$= -\frac{2}{3} + \frac{16}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos \frac{n\pi x}{2}$$

Deduction Part

Put $x = 0$ in the above equation, we get

$$f(0) = -\frac{2}{3} + \frac{16}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$

At $x = 0$, $f(x)$ is continuous

$$-2 = -\frac{2}{3} + \frac{16}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$

$$-\frac{4}{3} = \frac{16}{\pi^2} \left[-\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \dots \right]$$

$$-\frac{4}{3} = -\frac{16}{\pi^2} \left[\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots \right]$$

$$\left(-\frac{4}{3} \right) \left(-\frac{\pi^2}{16} \right) = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots$$

$$\therefore \frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots$$

$$\begin{aligned} f(x) &= x^2 - 2, \quad -2 < x < 2 \\ \text{Left hand limit} &= f(0-) = 2 \\ \text{Right hand limit} &= f(0+) = 2 \\ f(0-) &= f(0+) \text{ i.e., equal} \\ \therefore f(x) &\text{is continuous} \end{aligned}$$

Example: 3 Find the Fourier series expansion for $f(x) = kx$ in the interval $-l < x < l$

Solution: Let $f(x) = kx, -l < x < l$

$$f(-x) = k(-x) = -kx = -f(x)$$

$\therefore f(x)$ is an odd function of x . The Fourier series of $f(x)$ contains sine terms only, and hence $a_0 = a_n = 0$.

The Fourier series of $f(x)$ is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l} \quad \text{--- (1)}$$

where $b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$

To find b_n :

$$\begin{aligned} b_n &= \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx \\ &= \frac{2k}{l} \int_0^l x \sin \frac{n\pi x}{l} dx \\ &= \frac{2k}{l} \left\{ (x) \left(\frac{-\cos \frac{n\pi x}{l}}{\frac{n\pi}{l}} \right) - (1) \left(\frac{-\sin \frac{n\pi x}{l}}{\left(\frac{n\pi}{l} \right)^2} \right) \right\}_0^l \\ &= \frac{2k}{l} \left\{ -(x) \left(\frac{l}{n\pi} \right) \left(\cos \frac{n\pi x}{l} \right) + \left(\frac{l}{n\pi} \right)^2 \left(\sin \frac{n\pi x}{l} \right) \right\}_0^l \\ &= \frac{2k}{l} \left\{ \left(-\frac{l^2}{n\pi} (-1)^n + 0 \right) - (0 + 0) \right\} \{ \because \cos n\pi = (-1)^n; \sin n\pi = 0 \} \end{aligned}$$

$$b_n = \frac{2kl}{n\pi} (-1)^{n+1}$$

$\therefore$ (1) becomes

$$f(x) = \frac{2kl}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin \frac{n\pi x}{l}$$

Example: 4 Find the Fourier series for the function $f(x) = \begin{cases} 2, & -2 \leq x \leq 0 \\ x, & 0 \leq x \leq 2 \end{cases}$

Solution: Given function $f(x)$ is neither even nor odd function of x .

The Fourier series of $f(x)$ in the interval $-2 \leq x \leq 2$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{2} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{2} \quad \text{--- (1)}$$

Here $l = 2$

Where, $a_0 = \frac{1}{2} \int_{-2}^2 f(x) dx$

$$a_n = \frac{1}{2} \int_{-2}^2 f(x) \cos \frac{n\pi x}{2} dx \quad \& \quad b_n = \frac{1}{2} \int_{-2}^2 f(x) \sin \frac{n\pi x}{2} dx$$

To find a_0 :

$$\begin{aligned} a_0 &= \frac{1}{2} \int_{-2}^2 f(x) dx \\ &= \frac{1}{2} \left\{ \int_{-2}^0 f(x) dx + \int_0^2 f(x) dx \right\} \\ &= \frac{1}{2} \left\{ 2 \int_{-2}^0 dx + \int_0^2 x dx \right\} = \frac{1}{2} \left[2(x)_{-2}^0 + \left(\frac{x^2}{2} \right)_0^2 \right] = \frac{1}{2} [2(0 + 2) + (2 - 0)] \end{aligned}$$

$$a_0 = 3$$

To find a_n :

$$\begin{aligned} a_n &= \frac{1}{2} \int_{-2}^2 f(x) \cos \frac{n\pi x}{2} dx \\ &= \frac{1}{2} \left\{ \int_{-2}^0 2 \cos \frac{n\pi x}{2} dx + \int_0^2 x \cos \frac{n\pi x}{2} dx \right\} \\ &= \frac{1}{2} \left\{ \left[2 \left(\frac{\sin \frac{n\pi x}{2}}{\frac{n\pi}{2}} \right) \right]_{-2}^0 + \left[(x) \left(\frac{\sin \frac{n\pi x}{2}}{\frac{n\pi}{2}} \right) - (1) \left(\frac{-\cos \frac{n\pi x}{2}}{\left(\frac{n\pi}{2} \right)^2} \right) \right]_0^2 \right\} \\ &= \frac{1}{2} \left\{ \left[2 \left(\frac{2}{n\pi} \right) \left(\sin \frac{n\pi x}{2} \right) \right]_{-2}^0 + \left[(x) \left(\frac{2}{n\pi} \right) \left(\sin \frac{n\pi x}{2} \right) + (1) \left(\frac{2}{n\pi} \right)^2 \left(\cos \frac{n\pi x}{2} \right) \right]_0^2 \right\} \\ &= \frac{1}{2} \left\{ (0 - 0) + \left[\left(0 + \frac{4}{n^2 \pi^2} (-1)^n \right) - \left(0 + \frac{4}{n^2 \pi^2} \right) \right] \right\} \\ &= \frac{1}{2} \left\{ \frac{4}{n^2 \pi^2} (-1)^n - \frac{4}{n^2 \pi^2} \right\} \\ &= \frac{1}{2} \frac{4}{n^2 \pi^2} [(-1)^n - 1] = \frac{2}{n^2 \pi^2} [(-1)^n - 1] \end{aligned}$$

$$a_n = \begin{cases} \frac{-4}{n^2\pi^2} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

To find b_n :

$$\begin{aligned} b_n &= \frac{1}{2} \int_{-2}^2 f(x) \sin \frac{n\pi x}{2} dx \\ &= \frac{1}{2} \left\{ \int_{-2}^0 2 \sin \frac{n\pi x}{2} dx + \int_0^2 x \sin \frac{n\pi x}{2} dx \right\} \\ &= \frac{1}{2} \left\{ \left[2 \left(\frac{-\cos \frac{n\pi x}{2}}{\frac{n\pi}{2}} \right) \right]_{-2}^0 + \left[(x) \left(\frac{-\cos \frac{n\pi x}{2}}{\frac{n\pi}{2}} \right) - (1) \left(\frac{-\sin \frac{n\pi x}{2}}{\left(\frac{n\pi}{2} \right)^2} \right) \right]_0^2 \right\} \\ &= \frac{1}{2} \left\{ \left[-2 \left(\frac{2}{n\pi} \right) \left(\cos \frac{n\pi x}{2} \right) \right]_{-2}^0 + \left[-(x) \left(\frac{2}{n\pi} \right) \left(\cos \frac{n\pi x}{2} \right) + \left(\frac{2}{n\pi} \right)^2 \left(\sin \frac{n\pi x}{2} \right) \right]_0^2 \right\} \\ &= \frac{1}{2} \left\{ \left[\left(-\frac{4}{n\pi} \right) + \frac{4}{n\pi} (-1)^n \right] + \left[\left(-\frac{4}{n\pi} (-1)^n + 0 \right) - (0 + 0) \right] \right\} \\ &= \frac{1}{2} \left\{ -\frac{4}{n\pi} + \frac{4}{n\pi} (-1)^n - \frac{4}{n\pi} (-1)^n \right\} = \frac{1}{2} \left\{ -\frac{4}{n\pi} \right\} \end{aligned}$$

$$b_n = -\frac{2}{n\pi}$$

$$\therefore (1) \text{ becomes } f(x) = \frac{3}{2} - \frac{4}{\pi^2} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2} \cos \frac{n\pi x}{2} - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x}{2}$$

Practice problems

1. Find the Fourier series expansion of the periodic function $f(x)$ of period $2l$

defined by $f(x) = \begin{cases} l+x, & -l \leq x \leq 0 \\ l-x, & 0 \leq x \leq l \end{cases}$. Deduce that $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$

$$\text{Ans : } a_0 = l; a_n = \begin{cases} \frac{4l}{n^2\pi^2} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

2. Develop $f(x)$ in a Fourier series in the interval $(-2,2)$ if $f(x) = \begin{cases} 0, & -2 < x < 0 \\ 1, & 0 < x < 2 \end{cases}$

$$\text{Ans : } a_0 = 1; a_n = 0; b_n = \begin{cases} \frac{2}{n\pi} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

Fourier series in $(0, 2\pi)$

From equation (A), Put $l = \pi$, the interval $(0, 2l)$ becomes $(0, 2\pi)$. $\therefore$ The Fourier series of $f(x)$ in $(0, 2\pi)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$\text{Where, } a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx,$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx \quad \& \quad b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx$$

Fourier series in $(-\pi, \pi)$

From equation (B), Put $l = \pi$, the interval $(-l, l)$ becomes $(-\pi, \pi)$

$\therefore$ The Fourier series of $f(x)$ in $(-\pi, \pi)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$\text{where } a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \quad \& \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

Case: 1 Fourier series of Even Function in $(-\pi, \pi)$

If $f(x)$ is an even function of x in $(-\pi, \pi)$, then the Fourier expansion of $f(x)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$\text{where, } a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx \text{ \& } a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx$$

Case: 2 Fourier series of Odd Function in $(-\pi, \pi)$

If $f(x)$ is an odd function of x in $(-\pi, \pi)$, then the Fourier expansion of $f(x)$ is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

$$\text{where, } b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

ILLUSTRATIVE EXAMPLES UNDER $(0, 2\pi)$

Example: 1 Find the Fourier series of $f(x) = (\pi - x)^2$ in $(0, 2\pi)$ of period 2π and

hence deduce that $\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

Solution: Given $f(x) = (\pi - x)^2$ in $0 \leq x \leq 2\pi$

The Fourier series of $f(x)$ in $(0, 2\pi)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \quad \text{--- (1)}$$

$$\text{Where, } a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx,$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx \, dx \text{ \& } b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx$$

To find a_0 :

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_0^{2\pi} f(x) dx = \frac{1}{\pi} \int_0^{2\pi} (\pi - x)^2 dx \\ &= \frac{1}{\pi} \left\{ \frac{(\pi - x)^3}{-3} \right\}_0^{2\pi} \end{aligned}$$

$$= -\frac{1}{3\pi} \{(\pi - x)^3\}_0^{2\pi} = -\frac{1}{3\pi} \{-\pi^3 - \pi^3\} = -\frac{1}{3\pi} \{-2\pi^3\}$$

$$a_0 = \frac{2\pi^2}{3}$$

To find a_n :

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx \, dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} (\pi - x)^2 \cos nx \, dx$$

$$= \frac{1}{\pi} \left\{ \left[(\pi - x)^2 \left(\frac{\sin nx}{n} \right) - 2(\pi - x)(-1) \left(-\frac{\cos nx}{n^2} \right) + 2(1) \left(-\frac{\sin nx}{n^3} \right) \right] \right\}_0^{2\pi}$$

$$= \frac{1}{\pi} \left\{ \left[(\pi - x)^2 \left(\frac{\sin nx}{n} \right) - 2(\pi - x) \left(\frac{\cos nx}{n^2} \right) - \frac{2 \sin nx}{n^3} \right] \right\}_0^{2\pi}$$

$$= \frac{1}{\pi} \left\{ \left[0 - \frac{2(-\pi)}{n^2} - 0 \right] - \left[0 - \frac{2(\pi)}{n^2} - 0 \right] \right\} [\because \sin 2n\pi = 0 \text{ \& } \cos 2n\pi = 1]$$

$$= \frac{1}{\pi} \left\{ \frac{2\pi}{n^2} + \frac{2\pi}{n^2} \right\} = \frac{1}{\pi} \left[\frac{4\pi}{n^2} \right]$$

$$a_n = \frac{4}{n^2}$$

To find b_n :

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} (\pi - x)^2 \sin nx \, dx$$

$$= \frac{1}{\pi} \left\{ \left[(\pi - x)^2 \left(-\frac{\cos nx}{n} \right) - 2(\pi - x)(-1) \left(-\frac{\sin nx}{n^2} \right) + 2 \left(\frac{\cos nx}{n^3} \right) \right] \right\}_0^{2\pi}$$

$$\begin{aligned}
&= \frac{1}{\pi} \left\{ \left[-(\pi - x)^2 \left(\frac{\cos nx}{n} \right) - 2(\pi - x) \left(\frac{\sin nx}{n^2} \right) + 2 \left(\frac{\cos nx}{n^3} \right) \right] \right\}_0^{2\pi} \\
&= \frac{1}{\pi} \left\{ \left[-(-\pi)^2 \left(\frac{1}{n} \right) - 0 + 2 \left(\frac{1}{n^3} \right) \right] - \left[-\pi^2 - 0 + \frac{2}{n^3} \right] \right\} \\
&= \frac{1}{\pi} \left\{ \frac{-\pi^2}{n} + \frac{2}{n^3} + \frac{\pi^2}{n} - \frac{2}{n^3} \right\} [\because \sin 2n\pi = 0 \text{ \& } \cos 2n\pi = 1]
\end{aligned}$$

$$b_n = 0$$

$\therefore$ (1) becomes,

$$f(x) = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{1}{n^2} \cos nx$$

Deduction Part:

Put $x = 0$ in the above equation, we get

$$f(0) = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{1}{n^2} \quad \text{--- (2)}$$

At $x = 0$ the function $f(x)$ is continuous then (2) becomes

$$\begin{aligned}
\pi^2 &= \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{1}{n^2} \\
\therefore \frac{\pi^2}{6} &= \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots
\end{aligned}$$

Example: 2 Expand $f(x) = x(2\pi - x)$ as Fourier series in $(0, 2\pi)$ and hence deduce that the sum of the series of $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

Solution: Given $f(x) = x(2\pi - x) = 2\pi x - x^2$ in $0 \leq x \leq 2\pi$

The Fourier series of $f(x)$ in $(0, 2\pi)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \quad \text{--- (1)}$$

$$\text{here, } a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx,$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx \quad \& \quad b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx$$

To find a_0 :

$$\begin{aligned}
a_0 &= \frac{1}{\pi} \int_0^{2\pi} f(x) dx = \frac{1}{\pi} \int_0^{2\pi} (2\pi x - x^2) dx \\
&= \frac{1}{\pi} \left\{ \frac{2\pi x^2}{2} - \frac{x^3}{3} \right\}_0^{2\pi} = \frac{1}{\pi} \left\{ \pi x^2 - \frac{x^3}{3} \right\}_0^{2\pi} = \frac{1}{\pi} \left\{ \left(4\pi^3 - \frac{8\pi^3}{3} \right) - (0 - 0) \right\} \\
&= \frac{1}{\pi} \left\{ \frac{4\pi^3}{3} \right\} \\
a_0 &= \frac{4\pi^2}{3}
\end{aligned}$$

To find a_n :

$$\begin{aligned}
a_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx \\
&= \frac{1}{\pi} \int_0^{2\pi} (2\pi x - x^2) \cos nx dx \\
&= \frac{1}{\pi} \left\{ \left[(2\pi x - x^2) \left(\frac{\sin nx}{n} \right) - (2\pi - 2x) \left(-\frac{\cos nx}{n^2} \right) + (-2) \left(-\frac{\sin nx}{n^3} \right) \right] \right\}_0^{2\pi} \\
&= \frac{1}{\pi} \left\{ \left[0 + \frac{(2\pi - 4\pi)}{n^2} + 0 \right] - \left[0 + \frac{(2\pi - 0)}{n^2} + 0 \right] \right\} \\
&= \frac{1}{\pi} \left\{ -\frac{2\pi}{n^2} - \frac{2\pi}{n^2} \right\} = \frac{1}{\pi} \left\{ -\frac{4\pi}{n^2} \right\} [\because \sin 2n\pi = 0 \text{ \& } \cos 2n\pi = 1] \\
a_n &= -\frac{4}{n^2}
\end{aligned}$$

To find b_n :

$$\begin{aligned}
 b_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx \\
 &= \frac{1}{\pi} \int_0^{2\pi} (2\pi x - x^2) \sin nx \, dx \\
 &= \frac{1}{\pi} \left[\left[(2\pi x - x^2) \left(-\frac{\cos nx}{n} \right) - (2\pi - 2x) \left(-\frac{\sin nx}{n^2} \right) + (-2) \left(\frac{\cos nx}{n^3} \right) \right] \right]_0^{2\pi} \\
 &= \frac{1}{\pi} \left[\left[0 + 0 - \frac{2}{n^3} \right] - \left[0 + 0 - \frac{2}{n^3} \right] \right] \left[\because \sin 2n\pi = 0 \text{ \& } \cos 2n\pi = 1 \right] \\
 &= \frac{1}{\pi} \left\{ -\frac{2}{n^3} + \frac{2}{n^3} \right\}
 \end{aligned}$$

$$b_n = 0$$

$\therefore$ (1) becomes

$$f(x) = \frac{2\pi^2}{3} - 4 \sum_{n=1}^{\infty} \frac{1}{n^2} \cos nx \quad \text{--- (2)}$$

Deduction Part

At $x = 0$, the function $f(x)$ is continuous then (2) becomes

$$\begin{aligned}
 f(0) &= \frac{2\pi^2}{3} - 4 \sum_{n=1}^{\infty} \frac{1}{n^2} \\
 0 &= \frac{2\pi^2}{3} - 4 \sum_{n=1}^{\infty} \frac{1}{n^2} \\
 -\frac{2\pi^2}{3} &= -4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \dots \dots \right] \\
 \therefore \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \dots \dots &= \frac{\pi^2}{6}
 \end{aligned}$$

$\because f(x) = x(2\pi - x)$ in $0 < x < 2\pi$
 Left hand limit $f(0) = 0$
 Right hand limit $f(2\pi) = 0$
 Here, $f(0) = f(2\pi)$ i.e., equal
 $\therefore f(x)$ is continuous

Practice Problems

1. Obtain the Fourier series of $f(x) = x^2$; $0 < x < 2\pi$ and hence deduce that $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$

Ans: $a_0 = \frac{8\pi^2}{3}$; $a_n = \frac{4}{n^2}$; $b_n = -\frac{4\pi}{n}$; discontinuity appears at $x = 0$

2. Find the Fourier series of the function $f(x) = \begin{cases} 1, & 0 < x < \pi \\ 2, & \pi < x < 2\pi \end{cases}$

Ans: $a_0 = 3$; $a_n = 0$; $b_n = \begin{cases} -\frac{2}{\pi n} & ; \text{if } n \text{ is odd} \\ 0 & ; \text{if } n \text{ is even} \end{cases}$

3. If $f(x) = \left(\frac{\pi-x}{2}\right)^2$ in the interval $0 < x < 2\pi$, show that $f(x) = \frac{\pi^2}{12} + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$

ILLUSTRATIVE EXAMPLES UNDER $(-\pi, \pi)$

Example:1 Find the Fourier series expansion of $f(x) = \begin{cases} -x + 1, & -\pi \leq x \leq 0 \\ x + 1, & 0 \leq x \leq \pi \end{cases}$

Solution: Given $f(x) = \begin{cases} -x + 1, & -\pi \leq x \leq 0 \\ x + 1, & 0 \leq x \leq \pi \end{cases}$

Let $f_1(x) = -x + 1$; $f_2(x) = x + 1$

$f_1(-x) = -(-x) + 1 = x + 1 = f_2(x)$

$\therefore f(x)$ is an even function of x and hence the Fourier series of $f(x)$ contains cosine terms only, and hence $b_n = 0$.

The Fourier series of $f(x)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \quad \text{--- (1)}$$

where, $a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$ & $a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx$

To find a_0 :

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$\begin{aligned}
&= \frac{2}{\pi} \int_0^{\pi} (x+1) dx \quad \{\because f(x) = (x+1) \text{ in } 0 \leq x \leq \pi\} \\
&= \frac{2}{\pi} \left\{ \frac{x^2}{2} + x \right\}_0^{\pi} \\
&= \frac{2}{\pi} \left\{ \left(\frac{\pi^2}{2} + \pi \right) - (0) \right\} \\
&= \frac{2}{\pi} \left\{ \frac{\pi^2 + 2\pi}{2} \right\} = \frac{1}{\pi} \{\pi^2 + 2\pi\} = \frac{\pi(\pi+2)}{\pi}
\end{aligned}$$

$$a_0 = \pi + 2$$

To find a_n :

$$\begin{aligned}
a_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx \\
&= \frac{2}{\pi} \int_0^{\pi} (x+1) \cos nx dx \quad \{\because f(x) = (x+1) \text{ in } 0 \leq x \leq \pi\} \\
&= \frac{2}{\pi} \left\{ (x+1) \left(\frac{\sin nx}{n} \right) - (1) \left(-\frac{\cos nx}{n^2} \right) \right\}_0^{\pi} \\
&= \frac{2}{\pi} \left\{ \left(0 + \frac{\cos n\pi}{n^2} \right) - \left(0 + \frac{1}{n^2} \right) \right\} \{\because \cos n\pi = (-1)^n; \sin n\pi = 0\} \\
&= \frac{2}{\pi} \left\{ \left(0 + \frac{(-1)^n}{n^2} \right) - \left(0 + \frac{1}{n^2} \right) \right\} = \frac{2}{\pi n^2} [(-1)^n - 1]
\end{aligned}$$

$$a_n = \begin{cases} -\frac{4}{\pi n^2} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

Hence, (1) becomes

$$f(x) = \frac{(\pi+2)}{2} - \frac{4}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2} \cos nx$$

Deduction Part: Put $x = 0$ in the above equation, we get

$$f(0) = \frac{(\pi+2)}{2} - \frac{4}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2}$$

At $x = 0$, $f(x)$ is continuous

$$\begin{aligned}
1 &= \frac{(\pi+2)}{2} - \frac{4}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2} \\
\left(-\frac{\pi}{2}\right) \left(-\frac{\pi}{4}\right) &= \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \\
\therefore \frac{\pi^2}{8} &= \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots
\end{aligned}$$

IMPORTANT NOTE

If the trigonometric functions like **$\sin x, \cos x$** appears in the expansion of Fourier series, first we find the general term a_n and then to put $n = 0$ in the resulting function a_n , we can find the value of a_0 easily.

Example:2 Find the Fourier series expansion of $f(x) = x \sin x$ in $-\pi < x < \pi$.

Solution: Given $f(x) = x \sin x$; $-\pi < x < \pi$

$$f(-x) = (-x) \sin(-x) = x \sin x = f(x) \{\because \sin(-\theta) = -\sin \theta\}$$

$\therefore f(x) = x \sin x$ is an even function and hence $b_n = 0$

The Fourier series of $f(x)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \quad \text{--- (1)}$$

$$\text{where, } a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx \text{ \& } a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

To find a_n :

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \sin x \cos nx \, dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x [2 \cos nx \sin x] dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x [\sin(n+1)x - \sin(n-1)x] dx$$

$\therefore 2 \cos A \sin B = \sin(A+B) - \sin(A-B)$ Here $A = n$; $B = 1$

$$= \frac{1}{\pi} \left\{ \int_0^{\pi} x \sin(n+1)x \, dx - \int_0^{\pi} x \sin(n-1)x \, dx \right\}$$

$$= \frac{1}{\pi} \left\{ \left[(x) \left(\frac{-\cos(n+1)x}{(n+1)} \right) - (1) \left(\frac{-\sin(n+1)x}{(n+1)^2} \right) \right]_0^{\pi} \right. \\ \left. - \left[(x) \left(\frac{-\cos(n-1)x}{(n-1)} \right) - (1) \left(\frac{-\sin(n-1)x}{(n-1)^2} \right) \right]_0^{\pi} \right\}$$

$$= \frac{1}{\pi} \left\{ \left[- (x) \left(\frac{\cos(n+1)x}{(n+1)} \right) + \left(\frac{\sin(n+1)x}{(n+1)^2} \right) \right]_0^{\pi} \right. \\ \left. - \left[- (x) \left(\frac{\cos(n-1)x}{(n-1)} \right) + \left(\frac{\sin(n-1)x}{(n-1)^2} \right) \right]_0^{\pi} \right\}$$

$$= \frac{1}{\pi} \left\{ \left[\left(-(\pi) \frac{\cos(n+1)\pi}{(n+1)} + \frac{\sin(n+1)\pi}{(n+1)^2} \right) - (0+0) \right] \right. \\ \left. - \left[\left(-(\pi) \frac{\cos(n-1)\pi}{(n-1)} + \frac{\sin(n-1)\pi}{(n-1)^2} \right) - (0+0) \right] \right\}$$

$$= \frac{1}{\pi} \left\{ \frac{-\pi(-1)^{n+1}}{(n+1)} + \frac{\pi(-1)^{n+1}}{(n-1)} \right\}$$

$$= (-1)^{n+1} \left\{ \frac{-1}{(n+1)} + \frac{1}{(n-1)} \right\}$$

$$= (-1)^{n+1} \left\{ \frac{-n+1+n+1}{n^2-1} \right\}$$

$$a_n = (-1)^{n+1} \frac{2}{n^2-1} \text{ for } n \neq 1$$

Put $n = 0$ in the above, we get

$$a_0 = (-1)^{0+1} \frac{2}{0^2-1} = -\frac{2}{-1} = 2$$

$$a_0 = 2$$

Put $n = 1$ then $a_1 = \frac{2}{0} = \infty$

So the formula for a_n cannot be used to find a_1 . Hence, we have to find a_1 separately.

$$a_1 = \frac{2}{\pi} \int_0^{\pi} x \sin x \cos x \, dx = \frac{1}{\pi} \int_0^{\pi} x [2 \sin x \cos x] dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x \sin 2x \, dx \quad [\because \sin 2x = 2 \sin x \cos x]$$

$$= \frac{1}{\pi} \left\{ (x) \left(\frac{-\cos 2x}{2} \right) - (1) \left(\frac{-\sin 2x}{4} \right) \right\}_0^{\pi}$$

$$= \frac{1}{\pi} \left\{ \left(\frac{-\pi(1)}{2} + 0 \right) - (0+0) \right\}$$

$$a_1 = \frac{-1}{2}$$

Hence, (1) becomes

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx = \frac{a_0}{2} + a_1 \cos x + \sum_{n=2}^{\infty} a_n \cos nx$$

$$f(x) = 1 - \frac{\cos x}{2} + 2 \sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{(n^2-1)} \cos nx$$

Example: 3 Obtain Fourier series for the function $f(x) = |x|$ in $-\pi < x < \pi$ and

hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$ (OR)

A function $f(x)$ is defined as $f(x) = \begin{cases} -x, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$. Find the Fourier series of

$f(x)$ and hence deduce that $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$

Solution: Given $f(x) = |x|$ in $-\pi < x < \pi$

$$\text{i.e., } f(x) = \begin{cases} -x, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$$

$\therefore f(x) = |x|$ is an even function of x in $-\pi < x < \pi$ and Hence $b_n = 0$

The Fourier series of $f(x)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \quad \text{--- (1)}$$

$$\text{Where, } a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx, \quad a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

To find a_0 :

$$\begin{aligned} a_0 &= \frac{2}{\pi} \int_0^{\pi} f(x) dx \\ &= \frac{2}{\pi} \int_0^{\pi} x dx = \frac{2}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} \end{aligned}$$

$$a_0 = \pi$$

To find a_n :

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

$$\begin{aligned} &= \frac{2}{\pi} \int_0^{\pi} x \cos nx dx \\ &= \frac{2}{\pi} \left\{ \left[(x) \left(\frac{\sin nx}{n} \right) - (1) \left(\frac{-\cos nx}{n^2} \right) \right] \right\}_0^{\pi} \\ &= \frac{2}{\pi} \left\{ \left[0 + \frac{(-1)^n}{n^2} \right] - \left[0 + \frac{1}{n^2} \right] \right\} \\ &= \frac{2}{\pi} \left[\frac{(-1)^n}{n^2} - \frac{1}{n^2} \right] \\ &= \frac{2}{\pi n^2} [(-1)^n - 1] \end{aligned}$$

$$a_n = \begin{cases} \frac{-4}{\pi n^2}, & \text{when } n \text{ is odd} \\ 0, & \text{when } n \text{ is even} \end{cases}$$

Hence, (1) becomes,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$f(x) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2} \cos nx$$

Deduction Part: Put $x = 0$ in the above, we get

$$f(0) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2}$$

At $x = 0$, the function $f(x)$ is continuous

$$\begin{aligned} 0 &= \frac{\pi}{2} - \frac{4}{\pi} \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right) \\ -\frac{\pi}{2} &= -\frac{4}{\pi} \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right) \end{aligned}$$

$$\therefore \frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

$$f(x) = \begin{cases} -x, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$$

Left hand limit $f(0-) = 0$

Right hand limit $f(0+) = 0$

Here $f(0+) = f(0-)$ i.e., equal

$\therefore f(x)$ is continuous

Example : 4 Find the Fourier expansion of $x \cos x$ in the interval $-\pi < x < \pi$.

Solution : Given $f(x) = x \cos x$; $-\pi < x < \pi$

$$f(-x) = (-x) \cos(-x) = -x \cos x = -f(x) \{ \because \cos(-\theta) = \cos \theta \}$$

$\therefore f(x) = x \cos x$ is an odd function and hence $a_0 = a_n = 0$.

The Fourier series of $f(x)$ is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx \quad \text{--- (1)}$$

$$\text{where } b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx$$

To find b_n

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \cos x \sin nx dx = \frac{1}{\pi} \int_0^{\pi} x [2 \sin nx \cos x] dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x [\sin(n+1)x + \sin(n-1)x] dx$$

$$\because \sin A \cos B = \sin(A+B) + \sin(A-B)$$

$$= \frac{1}{\pi} \left\{ \int_0^{\pi} x \sin(n+1)x dx + \int_0^{\pi} x \sin(n-1)x dx \right\}$$

$$= \frac{1}{\pi} \left\{ \left[(x) \left(\frac{-\cos(n+1)x}{(n+1)} \right) - (1) \left(\frac{-\sin(n+1)x}{(n+1)^2} \right) \right]_0^{\pi} \right.$$

$$\left. + \left[(x) \left(\frac{-\cos(n-1)x}{(n-1)} \right) - (1) \left(\frac{-\sin(n-1)x}{(n-1)^2} \right) \right]_0^{\pi} \right\}$$

$$= \frac{1}{\pi} \left\{ \left[\left(-(\pi) \frac{\cos(n+1)\pi}{(n+1)} + \frac{\sin(n+1)\pi}{(n+1)^2} \right) - (0+0) \right] \right.$$

$$\left. + \left[\left(-(\pi) \frac{\cos(n-1)\pi}{(n-1)} + \frac{\sin(n-1)\pi}{(n-1)^2} \right) - (0+0) \right] \right\}$$

$$= \frac{1}{\pi} \left\{ \frac{-\pi(-1)^{n+1}}{(n+1)} - \frac{\pi(-1)^{n+1}}{(n-1)} \right\} \{ \because \cos(n \pm 1)\pi = (-1)^{n+1} \& \sin(n \pm 1)\pi = 0 \}$$

$$= -(-1)^{n+1} \left\{ \frac{1}{(n+1)} + \frac{1}{(n-1)} \right\}$$

$$= (-1)(-1)^n(-1)^1 \left\{ \frac{n-1+n+1}{n^2-1} \right\}$$

$$b_n = (-1)^n \frac{2n}{n^2-1} \quad \text{for } n \neq 1$$

$$\text{Put } n = 1, \text{ then } b_1 = -\frac{2}{0} = -\infty$$

So the formula for b_n cannot be used to find b_1 . Hence, we have to find b_1 separately.

$$b_1 = \frac{2}{\pi} \int_0^{\pi} x \cos x \sin x dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x [2 \sin x \cos x] dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x \sin 2x dx \{ \because \sin 2x = 2 \sin x \cos x \}$$

$$= \frac{1}{\pi} \left\{ (x) \left(\frac{-\cos 2x}{2} \right) - (1) \left(\frac{-\sin 2x}{4} \right) \right\}_0^{\pi}$$

$$= \frac{1}{\pi} \left\{ \left(\frac{-\pi(1)}{2} + 0 \right) - (0+0) \right\} \{ \because \cos 2\pi = 1; \sin 2\pi = 0 \}$$

$$b_1 = \frac{-1}{2}$$

$$\therefore (1) \text{ becomes, } f(x) = b_1 \sin x + \sum_{n=2}^{\infty} b_n \sin nx$$

$$f(x) = \frac{-1}{2} \sin x + 2 \sum_{n=2}^{\infty} \frac{n(-1)^n}{n^2-1} \sin nx$$

Example: 5 Find the Fourier expansion of $f(x) = x + x^2$ in the interval $-\pi < x < \pi$.

Solution: Given $f(x) = x + x^2$

$$f(-x) = -x + (-x)^2 = -x + x^2$$

The function $f(x)$ is neither even nor odd function of x .

The Fourier series of $f(x)$ in $(-\pi, \pi)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \quad \text{--- (1)}$$

where $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx \quad \& \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx$$

To find a_0 :

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} (x + x^2) dx \\ &= \frac{1}{\pi} \left\{ \int_{-\pi}^{\pi} x dx + \int_{-\pi}^{\pi} x^2 dx \right\} \\ &= \frac{1}{\pi} \left\{ 0 + 2 \int_0^{\pi} x^2 dx \right\} \left[\because \int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(x) \text{ is even} \\ 0 & , \text{if } f(x) \text{ is odd} \end{cases} \right] \\ &= \frac{2}{\pi} \left\{ \frac{x^3}{3} \right\}_0^{\pi} = \frac{2}{\pi} \left\{ \frac{\pi^3}{3} \right\} \\ a_0 &= \frac{2\pi^2}{3} \end{aligned}$$

To find a_n :

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} (x + x^2) \cos nx dx \\ &= \frac{1}{\pi} \left\{ \int_{-\pi}^{\pi} x \cos nx dx + \int_{-\pi}^{\pi} x^2 \cos nx dx \right\} \\ &= \frac{1}{\pi} \left\{ 0 + 2 \int_0^{\pi} x^2 \cos nx dx \right\} \left[\because \int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(x) \text{ is even} \\ 0 & , \text{if } f(x) \text{ is odd} \end{cases} \right] \\ &= \frac{2}{\pi} \left[(x^2) \left(\frac{\sin nx}{n} \right) - (2x) \left(-\frac{\cos nx}{n^2} \right) + (2) \left(-\frac{\sin nx}{n^3} \right) \right]_0^{\pi} \\ &= \frac{2}{\pi} \left[(x^2) \left(\frac{\sin nx}{n} \right) + (2x) \left(\frac{\cos nx}{n^2} \right) - (2) \left(\frac{\sin nx}{n^3} \right) \right]_0^{\pi} \\ &= \frac{2}{\pi} \left\{ \left[0 + \frac{(2\pi)(-1)^n}{n^2} - 0 \right] - [0 + 0 - 0] \right\} \left[\because \sin n\pi = 0 \text{ \& } \cos n\pi = (-1)^n \right] \\ a_n &= \frac{4(-1)^n}{n^2} \end{aligned}$$

To find b_n :

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} (x + x^2) \sin nx dx \\ &= \frac{1}{\pi} \left\{ \int_{-\pi}^{\pi} x \sin nx dx + \int_{-\pi}^{\pi} x^2 \sin nx dx \right\} \\ &= \frac{1}{\pi} \left\{ 2 \int_0^{\pi} x \sin nx dx + 0 \right\} \left[\because \int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx & \text{if } f(x) \text{ is even} \\ 0 & \text{if } f(x) \text{ is odd} \end{cases} \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{2}{\pi} \left\{ \left[(x) \left(-\frac{\cos nx}{n} \right) - (1) \left(-\frac{\sin nx}{n^2} \right) \right] \right\}_0^\pi \\
&= \frac{2}{\pi} \left\{ \left[-(x) \left(\frac{\cos nx}{n} \right) + \left(\frac{\sin nx}{n^2} \right) \right] \right\}_0^\pi \\
&= \frac{2}{\pi} \left\{ \left(\frac{-\pi(-1)^n}{n} + 0 \right) - (0 + 0) \right\} [\because \sin n\pi = 0 \text{ \& } \cos n\pi = (-1)^n]
\end{aligned}$$

$$b_n = \frac{2(-1)^{n+1}}{n}$$

$\therefore (1)$ becomes,

$$\begin{aligned}
f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \\
&= \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx + 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx
\end{aligned}$$

Practice problems

1. Find the Fourier series of $f(x) = x^2$; $-\pi < x < \pi$ and hence deduce that

$$(i) \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6} \quad (ii) \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$$

$$\text{Ans: } a_0 = \frac{2\pi^2}{3}; a_n = \frac{4(-1)^n}{n^2}$$

2. Find the Fourier series to represent $f(x) = |\sin x|$ in the interval $-\pi < x < \pi$.

$$\text{Ans: } a_n = \frac{-2}{(n+1)(n-1)\pi} [(-1)^n + 1], \text{ for } n \neq 1; a_0 = \frac{4}{\pi}; a_1 = 1$$

3. Expand $f(x) = |\cos x|$ as a Fourier series in the interval $-\pi < x < \pi$.

$$\text{Ans: } a_n = \begin{cases} 0, & \text{when } n \text{ is odd} \\ -\frac{4 \cos \frac{n\pi}{2}}{\pi(n^2-1)}, & \text{when } n \text{ is even} \end{cases}; \quad a_0 = \frac{4}{\pi}; \quad a_1 = 0$$

4. Determine the Fourier expansion of $f(x) = x$ in the interval $-\pi < x < \pi$.

$$\text{Ans: } b_n = \frac{-2(-1)^n}{n}$$

5. Expand $f(x)$ in a Fourier series where $f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$

Hence show that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$

$$\text{Ans: } a_0 = \frac{-\pi}{2}; a_n = \begin{cases} \frac{-2}{n^2} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}; b_n = \frac{(-1)^{n+1}}{n}$$

HALF RANGE SERIES

1. If the range is $0 < x < l$, then the half range cosine series for $f(x)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l}$$

$$\text{Where, } a_0 = \frac{2}{l} \int_0^l f(x) dx; \quad a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx$$

2. If the range is $0 < x < l$, then the half range sine series for $f(x)$ is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$$

$$\text{Where, } b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$$

REMARK: By putting $l = \pi$, the interval $0 < x < l$ becomes $0 < x < \pi$.

1. If the range is $0 < x < \pi$, then the half range cosine series for $f(x)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$\text{Where, } a_0 = \frac{2}{\pi} \int_0^\pi f(x) dx \quad \& \quad a_n = \frac{2}{\pi} \int_0^\pi f(x) \cos nx dx$$

2. If the range is $0 < x < \pi$, then the half range sine series for $f(x)$ is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

Where, $b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$

ILLUSTRATIVE EXAMPLES UNDER HALF-RANGE COSINE AND SINE SERIES

Example: 1 Obtain the Fourier cosine series of the function $f(x) =$

$$\begin{cases} kx, & 0 \leq x \leq \frac{l}{2} \\ k(l-x), & \frac{l}{2} \leq x \leq l \end{cases}$$

Solution: The half range cosine series for $f(x)$ in the interval $0 < x < l$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} \quad \text{--- (1)}$$

Where, $a_0 = \frac{2}{l} \int_0^l f(x) dx$ & $a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx$

To find a_0 :

$$\begin{aligned} a_0 &= \frac{2}{l} \int_0^l f(x) dx \\ &= \frac{2}{l} \left\{ \int_0^{\frac{l}{2}} f(x) dx + \int_{\frac{l}{2}}^l f(x) dx \right\} = \frac{2}{l} \left\{ k \int_0^{\frac{l}{2}} x dx + k \int_{\frac{l}{2}}^l (l-x) dx \right\} \\ &= \frac{2k}{l} \left\{ \left[\frac{x^2}{2} \right]_0^{\frac{l}{2}} + \left[\frac{(l-x)^2}{-2} \right]_{\frac{l}{2}}^l \right\} \left[\because \int x^2 dx = \frac{x^3}{3} \right] \\ &= \frac{2k}{l} \left\{ \frac{1}{2} [x^2]_0^{\frac{l}{2}} - \frac{1}{2} [(l-x)^2]_{\frac{l}{2}}^l \right\} \end{aligned}$$

$$\begin{aligned} &= \frac{2k}{l} \left\{ \frac{1}{2} \left[\frac{l^2}{4} - 0 \right] - \frac{1}{2} \left[0 - \frac{l^2}{4} \right] \right\} \\ &= \frac{2k}{l} \left\{ \frac{l^2}{8} + \frac{l^2}{8} \right\} \\ &= \frac{2k}{l} \left\{ \frac{2l^2}{8} \right\} \end{aligned}$$

$$a_0 = \frac{kl}{2}$$

To find a_n :

$$\begin{aligned} a_n &= \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx = \frac{2}{l} \left\{ \int_0^{\frac{l}{2}} f(x) \cos \frac{n\pi x}{l} dx + \int_{\frac{l}{2}}^l f(x) \cos \frac{n\pi x}{l} dx \right\} \\ &= \frac{2}{l} \left\{ k \int_0^{\frac{l}{2}} x \cos \frac{n\pi x}{l} dx + k \int_{\frac{l}{2}}^l (l-x) \cos \frac{n\pi x}{l} dx \right\} \\ &= \frac{2k}{l} \left\{ \left[(x) \left(\frac{\sin \frac{n\pi x}{l}}{\frac{n\pi}{l}} \right) - (1) \left(\frac{-\cos \frac{n\pi x}{l}}{\frac{n^2 \pi^2}{l^2}} \right) \right]_0^{\frac{l}{2}} \right. \\ &\quad \left. + \left[(l-x) \left(\frac{\sin \frac{n\pi x}{l}}{\frac{n\pi}{l}} \right) - (-1) \left(\frac{-\cos \frac{n\pi x}{l}}{\frac{n^2 \pi^2}{l^2}} \right) \right]_{\frac{l}{2}}^l \right\} \\ &= \frac{2k}{l} \left\{ \left[(x) \left(\frac{l}{n\pi} \right) \left(\sin \frac{n\pi x}{l} \right) + \left(\frac{l^2}{n^2 \pi^2} \right) \left(\cos \frac{n\pi x}{l} \right) \right]_0^{\frac{l}{2}} \right. \\ &\quad \left. + \left[(l-x) \left(\frac{l}{n\pi} \right) \left(\sin \frac{n\pi x}{l} \right) - \left(\frac{l^2}{n^2 \pi^2} \right) \left(\cos \frac{n\pi x}{l} \right) \right]_{\frac{l}{2}}^l \right\} \end{aligned}$$

$$\begin{aligned}
&= \frac{2k}{l} \left\{ \left[\left(\frac{l}{2} \right) \left(\frac{l}{n\pi} \right) \left(\sin \frac{n\pi}{2} \right) + \left(\frac{l^2}{n^2\pi^2} \right) \left(\cos \frac{n\pi}{2} \right) \right] - \left[0 + \left(\frac{l^2}{n^2\pi^2} \right) \right] \right\} \\
&\quad + \left\{ \left[0 - \frac{l^2}{n^2\pi^2} (-1)^n \right] - \left[\left(\frac{l}{2} \right) \left(\frac{l}{n\pi} \right) \left(\sin \frac{n\pi}{2} \right) - \left(\frac{l^2}{n^2\pi^2} \right) \left(\cos \frac{n\pi}{2} \right) \right] \right\} \\
&= \frac{2k}{l} \left\{ \left(\frac{l}{2} \right) \left(\frac{l}{n\pi} \right) \left(\sin \frac{n\pi}{2} \right) + \left(\frac{l^2}{n^2\pi^2} \right) \left(\cos \frac{n\pi}{2} \right) - \left(\frac{l^2}{n^2\pi^2} \right) - \frac{l^2}{n^2\pi^2} (-1)^n \right. \\
&\quad \left. - \left(\frac{l}{2} \right) \left(\frac{l}{n\pi} \right) \left(\sin \frac{n\pi}{2} \right) + \left(\frac{l^2}{n^2\pi^2} \right) \left(\cos \frac{n\pi}{2} \right) \right\} \\
&= \frac{2k}{l} \left\{ 2 \left(\frac{l^2}{n^2\pi^2} \right) \left(\cos \frac{n\pi}{2} \right) - \left(\frac{l^2}{n^2\pi^2} \right) - \left(\frac{l^2}{n^2\pi^2} \right) (-1)^n \right\} \\
&= \frac{2k}{l} \left\{ 2 \left(\frac{l^2}{n^2\pi^2} \right) \left(\cos \frac{n\pi}{2} \right) - \left(\frac{l^2}{n^2\pi^2} \right) [1 + (-1)^n] \right\}
\end{aligned}$$

$$a_n = \frac{2kl}{n^2\pi^2} \left\{ 2 \left(\cos \frac{n\pi}{2} \right) - [1 + (-1)^n] \right\}$$

$\therefore (1)$ becomes,

$$f(x) = \frac{kl}{4} + \frac{2kl}{\pi^2} \sum_{n=1}^{\infty} \left(\frac{1}{n^2} \right) \left\{ 2 \left(\cos \frac{n\pi}{2} \right) - [1 + (-1)^n] \right\} \cos \frac{n\pi x}{l}$$

Example: 2 Obtain the sine series of the function $f(x) = \begin{cases} cx, & 0 \leq x \leq \frac{l}{2} \\ c(l-x), & \frac{l}{2} \leq x \leq l \end{cases}$

Solution: The half range sine series for $f(x)$ is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l} \quad \text{--- (1)}$$

Where, $b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$

To find b_n :

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$$

$$\begin{aligned}
&= \frac{2}{l} \left\{ \int_0^{\frac{l}{2}} f(x) \sin \frac{n\pi x}{l} dx + \int_{\frac{l}{2}}^l f(x) \sin \frac{n\pi x}{l} dx \right\} \\
&= \frac{2}{l} \left\{ c \int_0^{\frac{l}{2}} x \sin \frac{n\pi x}{l} dx + c \int_{\frac{l}{2}}^l (l-x) \sin \frac{n\pi x}{l} dx \right\} \\
&= \frac{2c}{l} \left\{ \left[(x) \left(\frac{-\cos \frac{n\pi x}{l}}{\frac{n\pi}{l}} \right) - (1) \left(\frac{-\sin \frac{n\pi x}{l}}{\frac{n^2\pi^2}{l^2}} \right) \right]_0^{\frac{l}{2}} \right. \\
&\quad \left. + \left[(l-x) \left(\frac{-\cos \frac{n\pi x}{l}}{\frac{n\pi}{l}} \right) - (-1) \left(\frac{-\sin \frac{n\pi x}{l}}{\frac{n^2\pi^2}{l^2}} \right) \right]_{\frac{l}{2}}^l \right\} \\
&= \frac{2c}{l} \left\{ \left[- (x) \left(\frac{l}{n\pi} \right) \left(\cos \frac{n\pi x}{l} \right) + \left(\frac{l^2}{n^2\pi^2} \right) \left(\sin \frac{n\pi x}{l} \right) \right]_0^{\frac{l}{2}} \right. \\
&\quad \left. + \left[- (l-x) \left(\frac{l}{n\pi} \right) \left(\cos \frac{n\pi x}{l} \right) - \left(\frac{l^2}{n^2\pi^2} \right) \left(\sin \frac{n\pi x}{l} \right) \right]_{\frac{l}{2}}^l \right\} \\
&= \frac{2c}{l} \left\{ \left[\left(- \left(\frac{l}{2} \right) \left(\frac{l}{n\pi} \right) \left(\cos \frac{n\pi}{2} \right) + \left(\frac{l^2}{n^2\pi^2} \right) \left(\sin \frac{n\pi}{2} \right) \right] - [0] \right) \right. \\
&\quad \left. + \left[[0 - 0] - \left[- \left(\frac{l}{2} \right) \left(\frac{l}{n\pi} \right) \left(\cos \frac{n\pi}{2} \right) - \left(\frac{l^2}{n^2\pi^2} \right) \left(\sin \frac{n\pi}{2} \right) \right] \right] \right\}
\end{aligned}$$

$$= \frac{2c}{l} \left[\frac{2l^2}{n^2\pi^2} \sin \frac{n\pi}{2} \right]$$

$$b_n = \frac{4cl}{n^2\pi^2} \sin \left(\frac{n\pi}{2} \right)$$

Hence, (1) becomes

$$f(x) = \frac{4cl}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \left(\frac{n\pi}{2} \right) \sin \left(\frac{n\pi x}{l} \right)$$

Example: 3 Find the half range sine series of $f(x) = lx - x^2$ in $(0, l)$

Solution: The half range sine series for $f(x)$ is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l} \quad \text{--- (1)}$$

Where, $b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$

To find b_n :

$$\begin{aligned} b_n &= \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx \\ &= \frac{2}{l} \int_0^l (lx - x^2) \sin \frac{n\pi x}{l} dx \\ &= \frac{2}{l} \left\{ (lx - x^2) \left(\frac{-\cos \frac{n\pi x}{l}}{\frac{n\pi}{l}} \right) - (l - 2x) \left(\frac{-\sin \frac{n\pi x}{l}}{\frac{n^2 \pi^2}{l^2}} \right) + (-2) \left(\frac{\cos \frac{n\pi x}{l}}{\frac{n^3 \pi^3}{l^3}} \right) \right\}_0^l \\ &= \frac{2}{l} \left\{ -(lx - x^2) \left(\frac{l}{n\pi} \right) \left(\cos \frac{n\pi x}{l} \right) + (l - 2x) \left(\frac{l^2}{n^2 \pi^2} \right) \left(\sin \frac{n\pi x}{l} \right) \right. \\ &\quad \left. - (2) \left(\frac{l^3}{n^3 \pi^3} \right) \left(\cos \frac{n\pi x}{l} \right) \right\}_0^l \\ &= \frac{2}{l} \left\{ \left[0 + 0 - 2 \left(\frac{l^3}{n^3 \pi^3} \right) (-1)^n \right] - \left[0 + 0 - 2 \left(\frac{l^3}{n^3 \pi^3} \right) \right] \right\} \\ &= \frac{2}{l} \left\{ -2 \left(\frac{l^3}{n^3 \pi^3} \right) (-1)^n + 2 \left(\frac{l^3}{n^3 \pi^3} \right) \right\} \\ &= \frac{2}{l} \cdot 2 \left(\frac{l^3}{n^3 \pi^3} \right) [1 - (-1)^n] = \frac{4l^2}{n^3 \pi^3} [1 - (-1)^n] \end{aligned}$$

$$b_n = \begin{cases} \frac{8l^2}{n^3 \pi^3} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

Hence, (1) becomes

$$f(x) = \frac{8l^2}{\pi^3} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^3} \sin \frac{n\pi x}{l}$$

Example: 4 Find the half range cosine series for $f(x) = (\pi - x)^2$ in $0 < x < \pi$.

Solution: Given $f(x) = (\pi - x)^2$. Half range cosine series is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \quad \text{--- (1)}$$

Where, $a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$, $a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$

To find a_0 :

$$\begin{aligned} a_0 &= \frac{2}{\pi} \int_0^{\pi} f(x) dx \\ &= \frac{2}{\pi} \int_0^{\pi} (\pi - x)^2 dx \\ &= \frac{2}{\pi} \left[\frac{(\pi - x)^3}{-3} \right]_0^{\pi} \left\{ \because \int x^2 dx = \frac{x^3}{3} \right\} \\ &= -\frac{2}{3\pi} [(\pi - x)^3]_0^{\pi} = -\frac{2}{3\pi} [0 - \pi^3] \end{aligned}$$

$$a_0 = \frac{2\pi^2}{3}$$

To find a_n

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} (\pi - x)^2 \cos nx dx \\ &= \frac{2}{\pi} \left\{ \left[(\pi - x)^2 \left(\frac{\sin nx}{n} \right) - 2(\pi - x)(-1) \left(\frac{-\cos nx}{n^2} \right) + 2(-1) \left(\frac{-\sin nx}{n^3} \right) \right]_0^{\pi} \right\} \end{aligned}$$

$$= \frac{2}{\pi} \left\{ \left[(\pi - x)^2 \left(\frac{\sin nx}{n} \right) - 2(\pi - x) \left(\frac{\cos nx}{n^2} \right) + 2 \left(\frac{\sin nx}{n^3} \right) \right]_0^{\pi} \right\}$$

$$= \frac{2}{\pi} \left\{ [0 + 0 + 0] - \left[0 - 2\pi \left(\frac{1}{n^2} \right) + 0 \right] \right\} = \frac{2}{\pi} \left\{ \frac{2\pi}{n^2} \right\}$$

$$a_n = \frac{4}{n^2}$$

Hence, (1) becomes

$$f(x) = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{1}{n^2} \cos nx$$

Example: 5 Obtain the half range cosine series for $f(x) = (x - 1)^2$ in the interval

$0 < x < 1$. Hence deduce that $\frac{\pi^2}{24} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

Solution: Given $f(x) = (x - 1)^2, 0 < x < 1$ Here $l = 1$

The half range cosine series in the interval $0 < x < l$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \quad \text{--- (1)}$$

$$\text{Where, } a_0 = \frac{2}{l} \int_0^l f(x) dx \quad \& a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx$$

To find a_0 :

$$\begin{aligned} a_0 &= \frac{2}{l} \int_0^l f(x) dx \\ &= \frac{2}{1} \int_0^1 (x - 1)^2 dx \quad \{\because l = 1\} \\ &= 2 \left[\frac{(x - 1)^3}{3} \right]_0^1 \left\{ \because \int x^2 dx = \frac{x^3}{3} \right\} \\ &= \frac{2}{3} [(x - 1)^3]_0^1 = \frac{2}{3} [0 - (-1)] \end{aligned}$$

$$a_0 = \frac{2}{3}$$

To find a_n :

$$\begin{aligned} a_n &= \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx = \frac{2}{1} \int_0^1 f(x) \cos n\pi x dx \quad \{\because l = 1\} \\ &= 2 \int_0^1 (x - 1)^2 \cos n\pi x dx \\ &= 2 \left\{ (x - 1)^2 \left(\frac{\sin n\pi x}{n\pi} \right) - 2(x - 1) \left(-\frac{\cos n\pi x}{n^2\pi^2} \right) + 2(1) \left(-\frac{\sin n\pi x}{n^3\pi^3} \right) \right\}_0^1 \\ &= 2 \left\{ [0 + 0 + 0] - \left[0 - 2 \left(\frac{1}{n^2\pi^2} \right) + 0 \right] \right\} \end{aligned}$$

$$a_n = \frac{4}{n^2\pi^2}$$

$\therefore$ (1) becomes,

$$f(x) = \frac{1}{2} \left(\frac{2}{3} \right) + \sum_{n=1}^{\infty} \frac{4}{n^2\pi^2} \cos \frac{n\pi x}{l}$$

$$f(x) = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \cos n\pi x$$

Deduction Part

At $x = 0$, the function $f(x)$ is discontinuous

$$\left(\frac{1 + 0}{2} \right) = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$\frac{1}{6} = \frac{4}{\pi^2} \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right]$$

$$\therefore \frac{\pi^2}{24} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

$$f(x) = (x - 1)^2 \text{ in } 0 < x < 1$$

Left hand limit $f(0) = 1$

Right hand limit $f(1) = 0$

$f(0) \neq f(1)$ i.e. unequal

$\therefore f(x)$ is discontinuous

Practice problems

1. Obtain the half range cosine series for $f(x) = x$ in the interval $(0, \pi)$.

$$\text{Ans: } a_0 = \pi; a_n = \begin{cases} 0, & \text{when } n \text{ is even} \\ -\frac{4}{n^2\pi}, & \text{when } n \text{ is odd} \end{cases};$$

2. Obtain the half range sine series of $f(x) = x(\pi - x)$ in the interval $(0, \pi)$. and

$$\text{deduce that } \frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \dots \infty = \frac{\pi^3}{32}$$

$$\text{Ans: } b_n = \begin{cases} 0, & \text{when } n \text{ is even} \\ \frac{8}{n^3\pi}, & \text{when } n \text{ is odd} \end{cases}$$

3. Obtain the half range sine series for $f(x) = x$ in the interval $(0, l)$ and hence

$$\text{deduce that } \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

$$\text{Ans: } b_n = \frac{2l}{n\pi} (-1)^{n+1}$$

4. Obtain the Fourier cosine series expansion of $x \sin x$ in $(0, \pi)$ and hence find

$$\text{the value of } \frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \dots = \frac{\pi-2}{4}$$

$$\text{Ans: } a_0 = 2; a_n = \frac{2(-1)^{n+1}}{n^2} \text{ for } n \neq 1; a_1 = -\frac{1}{2}$$

ROOT MEAN SQUARE VALUE(R.M.S)

The R.M.S value of function $y = f(x)$ defined over the interval (a, b) is given by

$$\bar{y} = \sqrt{\frac{1}{b-a} \int_a^b y^2 dx}$$

Parseval's identity

1. If the half range cosine series of $f(x)$ in $(0, l)$ is $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l}$,

$$\text{then } \frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} a_n^2 = \frac{1}{l} \int_0^l [f(x)]^2 dx$$

2. If the half range sine series of $f(x)$ in $(0, l)$ is $f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$, then

$$\frac{1}{2} \sum_{n=1}^{\infty} b_n^2 = \frac{1}{l} \int_0^l [f(x)]^2 dx$$

Put $l = \pi$ in the above formulas, we get

3. If the half range cosine series of $f(x)$ in $(0, \pi)$ is $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$, then

$$\frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} a_n^2 = \frac{1}{\pi} \int_0^{\pi} [f(x)]^2 dx$$

4. If the half range sine series of $f(x)$ in $(0, \pi)$ is $f(x) = \sum_{n=1}^{\infty} b_n \sin nx$, then

$$\frac{1}{2} \sum_{n=1}^{\infty} b_n^2 = \frac{1}{\pi} \int_0^{\pi} [f(x)]^2 dx$$

ILLUSTRATIVE EXAMPLES UNDER ROOT MEAN SQUARE VALUE FUNCTION

Example: 1 Find the root mean square value of the function $f(x) = x$ in the interval $(0, l)$.

Solution: The r.m.s value of $y = f(x)$ in $(0, l)$

$$\begin{aligned} \bar{y} &= \sqrt{\frac{1}{b-a} \int_a^b y^2 dx} \\ &= \sqrt{\frac{1}{l-0} \int_0^l x^2 dx} \text{ \{here. } a=0, b=l\} \\ &= \sqrt{\frac{1}{l} \left(\frac{x^3}{3} \right)_0^l} = \sqrt{\frac{1}{l} \left(\frac{l^3}{3} \right)} = \sqrt{\frac{l^2}{3}} \\ \therefore \bar{y} &= \frac{l}{\sqrt{3}} \end{aligned}$$

Example: 2 Find the root mean square value of the function $f(x) = x^2$ in the interval $(0, l)$.

Solution: The r.m.s value of $y = f(x)$ in $(0, l)$

$$\begin{aligned}\bar{y} &= \sqrt{\frac{1}{b-a} \int_a^b y^2 dx} \\ &= \sqrt{\frac{1}{l-0} \int_0^l (x^2)^2 dx} \text{ Here } a=0, b=l \\ &= \sqrt{\frac{1}{l} \int_0^l x^4 dx} = \sqrt{\frac{1}{l} \left(\frac{x^5}{5} \right)_0^l} = \sqrt{\frac{1}{l} \left(\frac{l^5}{5} \right)} = \sqrt{\frac{l^4}{5}} \\ \therefore \bar{y} &= \frac{l^2}{\sqrt{5}}\end{aligned}$$

Example: 3 State Parseval's identity for full range expansion of $f(x)$ as Fourier series in $(0, 2l)$.

Solution:

$$\begin{aligned}\frac{1}{2l} \int_0^{2l} [f(x)]^2 dx &= \frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \text{ (or)} \\ \frac{1}{l} \int_0^{2l} [f(x)]^2 dx &= \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)\end{aligned}$$

Practice problems

1. Find the root mean square value of the function $f(x) = x$ in the interval $(0, \pi)$.

$$\text{Ans: } \bar{y} = \frac{\pi}{\sqrt{3}}$$

2. Find the root mean square value of the function $f(x) = x^2$ in the interval $(0, \pi)$

$$\text{Ans: } \bar{y} = \frac{\pi^2}{\sqrt{5}}$$

ILLUSTRATIVE EXAMPLES

Example:1 Find the Fourier series for $f(x) = x^2$ in $(-\pi, \pi)$. Hence find

$$\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \dots \dots \dots \infty$$

Solution: Given $f(x) = x^2$ in $(-\pi, \pi)$

$$f(-x) = (-x)^2 = x^2 = f(x)$$

$f(x)$ is an even function of x .

$\therefore$ The Fourier series of $f(x)$ contains cosine terms only, and hence $b_n = 0$.

The Fourier series of $f(x)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \quad \text{--- (1)}$$

$$\text{where, } a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx \quad \& \quad a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

To find a_0 :

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} x^2 dx = \frac{2}{\pi} \left(\frac{x^3}{3} \right)_0^{\pi}$$

$$a_0 = \frac{2\pi^2}{3}$$

To find a_n

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx dx$$

$$= \frac{2}{\pi} \left\{ \left[(x^2) \left(\frac{\sin nx}{n} \right) - (2x) \left(\frac{-\cos nx}{n^2} \right) + 2 \left(\frac{-\sin nx}{n^3} \right) \right]_0^{\pi} \right\}$$

$$= \frac{2}{\pi} \left\{ \left[0 + \frac{2\pi(-1)^n}{n^2} + 0 \right] - [0 + 0 + 0] \right\} \left\{ \because \sin n\pi = 0; \cos n\pi = (-1)^n \right\}$$

$$a_n = \frac{4(-1)^n}{n^2}$$

Hence, (1) becomes

$$f(x) = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx$$

Deduction Part:[using Parseval's identity]

The Parseval's theorem for half range cosine series in the interval $(0, \pi)$ is

$$\frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} a_n^2 = \frac{1}{\pi} \int_0^{\pi} [f(x)]^2 dx$$

$$\text{Since } f(x) = x^2 \text{ in } (0, \pi); a_0 = \frac{2\pi^2}{3}; a_n = \frac{4(-1)^n}{n^2}$$

$$\frac{1}{4} \cdot \frac{4\pi^4}{9} + \frac{1}{2} \sum_{n=1}^{\infty} \frac{16(-1)^{2n}}{n^4} = \frac{1}{\pi} \int_0^{\pi} x^4 dx$$

$$\frac{\pi^4}{9} + 8 \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{1}{\pi} \left\{ \frac{x^5}{5} \right\}_0^{\pi} \{ \because (-1)^{2n} = 1 \}$$

$$\frac{\pi^4}{9} + 8 \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{1}{\pi} \frac{\pi^5}{5}$$

$$8 \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{5} - \frac{\pi^4}{9} = \frac{4\pi^4}{45}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{4\pi^4}{45} \times \frac{1}{8}$$

$$\therefore \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots \dots \infty = \frac{\pi^4}{90}$$

Example: 2 Expand $(x) = \pi x - x^2, 0 < x < \pi$ as a Fourier cosine series and

Hence find $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$.

Solution: Given $(x) = \pi x - x^2, 0 < x < \pi$

The half-range cosine series for $f(x)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \quad \text{--- (1)}$$

$$\text{where, } a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx \text{ \& } a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

To find a_0 :

$$\begin{aligned} a_0 &= \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} (\pi x - x^2) dx \\ &= \frac{2}{\pi} \left\{ \frac{\pi x^2}{2} - \frac{x^3}{3} \right\}_0^{\pi} = \frac{2}{\pi} \left\{ \left[\frac{\pi^3}{2} - \frac{\pi^3}{3} \right] \right\} = \frac{2}{\pi} \left\{ \frac{\pi^3}{6} \right\} \end{aligned}$$

$$a_0 = \frac{\pi^2}{3}$$

To find a_n

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} (\pi x - x^2) \cos nx dx \\ &= \frac{2}{\pi} \left\{ \left[(\pi x - x^2) \left(\frac{\sin nx}{n} \right) - (\pi - 2x) \left(\frac{-\cos nx}{n^2} \right) + (-2) \left(\frac{-\sin nx}{n^3} \right) \right]_0^{\pi} \right\} \\ &= \frac{2}{\pi} \left\{ \left[0 + \frac{(-\pi)(-1)^n}{n^2} + 0 \right] - \left[0 + \frac{\pi}{n^2} + 0 \right] \right\} [\because \sin n\pi = 0; \cos n\pi = (-1)^n] \\ &= \frac{-2}{n^2} \{ (-1)^n + 1 \} \end{aligned}$$

$$a_n = \begin{cases} \frac{-4}{n^2}, & \text{if } n \text{ is even} \\ 0 & \text{if } n \text{ is odd} \end{cases}$$

Hence, (1) becomes

$$f(x) = \frac{\pi^2}{6} - 4 \sum_{n=2,4,6,\dots}^{\infty} \frac{1}{n^2} \cos nx$$

DEDUCTION PART:[using Parseval's identity]

The Parseval's identity for half range cosine series in the interval $(0, \pi)$ is

$$\frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} a_n^2 = \frac{1}{\pi} \int_0^{\pi} [f(x)]^2 dx$$

Since $f(x) = \pi x - x^2$ in $(0, \pi)$; $a_0 = \frac{\pi^2}{3}$; $a_n = \begin{cases} -4 & \text{if } n \text{ is even} \\ 0 & \text{if } n \text{ is odd} \end{cases}$

$$\frac{1}{4} \cdot \frac{\pi^4}{9} + \frac{1}{2} \sum_{n=2,4,6,\dots}^{\infty} \frac{16}{n^4} = \frac{1}{\pi} \int_0^{\pi} [\pi x - x^2]^2 dx \quad \text{--- (2)}$$

$$= \frac{1}{\pi} \int_0^{\pi} [\pi^2 x^2 - 2\pi x^3 + x^4] dx$$

$$= \frac{1}{\pi} \left\{ \pi^2 \frac{x^3}{3} - 2\pi \frac{x^4}{4} + \frac{x^5}{5} \right\}_0^{\pi}$$

$$= \frac{1}{\pi} \left\{ \frac{\pi^5}{3} - \frac{\pi^5}{2} + \frac{\pi^5}{5} \right\}$$

$$= \frac{1}{\pi} \left\{ -\frac{\pi^5}{6} + \frac{\pi^5}{5} \right\} = \frac{1}{\pi} \left\{ \frac{\pi^5}{30} \right\}$$

$$\therefore (1) \text{ becomes } \frac{1}{4} \cdot \frac{\pi^4}{9} + \frac{1}{2} \sum_{n=2,4,6,\dots}^{\infty} \frac{16}{n^4} = \left\{ \frac{\pi^4}{30} \right\}$$

$$\frac{1}{2} \sum_{n=2,4,6,\dots}^{\infty} \frac{16}{n^4} = \frac{\pi^4}{30} - \frac{\pi^4}{36}$$

$$\frac{1}{2} \sum_{n=1}^{\infty} \frac{16}{(2n)^4} = \frac{\pi^4}{180} \{ \because n \text{ is even, Replace } n = 2n \}$$

$$\frac{16}{32} \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{180}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{180} \times 2 = \frac{\pi^4}{90}$$

Example: 3 Find the half range cosine series for $f(x) = (\pi - x)^2$ in $0 < x < \pi$.

Hence find the sum of the series $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots \dots \infty = \frac{\pi^4}{90}$

Solution: Given $f(x) = (\pi - x)^2$

Half range cosine series is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \quad \text{--- (2)}$$

$$\text{Where, } a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx \text{ \& } a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

To find a_0 :

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} (\pi - x)^2 dx$$

$$= \frac{2}{\pi} \left[\frac{(\pi - x)^3}{-3} \right]_0^{\pi} \left\{ \because \int x^2 dx = \frac{x^3}{3} \right\}$$

$$= -\frac{2}{3\pi} [(\pi - x)^3]_0^{\pi} = -\frac{2}{3\pi} [0 - \pi^3]$$

$$a_0 = \frac{2\pi^2}{3}$$

To find a_n :

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

$$= \frac{2}{\pi} \int_0^{\pi} (\pi - x)^2 \cos nx dx$$

$$= \frac{2}{\pi} \left\{ \left[(\pi - x)^2 \left(\frac{\sin nx}{n} \right) - 2(\pi - x)(-1) \left(\frac{-\cos nx}{n^2} \right) + 2(-1) \left(\frac{-\sin nx}{n^3} \right) \right]_0^{\pi} \right\}$$

$$= \frac{2}{\pi} \left\{ \left[(\pi - x)^2 \left(\frac{\sin nx}{n} \right) - 2(\pi - x) \left(\frac{\cos nx}{n^2} \right) + 2 \left(\frac{\sin nx}{n^3} \right) \right]_0^{\pi} \right\}$$

$$= \frac{2}{\pi} \left\{ [0 + 0 + 0] - \left[0 - 2\pi \left(\frac{1}{n^2} \right) + 0 \right] \right\} = \frac{2}{\pi} \left\{ \frac{2\pi}{n^2} \right\}$$

$$a_n = \frac{4}{n^2}$$

$$\therefore (1) \text{ becomes, } f(x) = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{1}{n^2} \cos nx$$

$$\text{We have } a_0 = \frac{2\pi^2}{3}; a_n = \frac{4}{n^2}; \text{ also, } f(x) = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{1}{n^2} \cos nx$$

The Parseval's identity for half range cosine series in the interval $(0, \pi)$ is

$$\frac{a_0^2}{4} + \frac{1}{2} \sum_{n=1}^{\infty} a_n^2 = \frac{1}{\pi} \int_0^{\pi} [f(x)]^2 dx$$

$$\frac{1}{4} \left(\frac{2\pi^2}{3} \right)^2 + \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{4}{n^2} \right)^2 = \frac{1}{\pi} \int_0^{\pi} [(\pi - x)^2]^2 dx$$

$$\frac{1}{4} \left(\frac{4\pi^4}{9} \right) + \frac{1}{2} \sum_{n=1}^{\infty} \frac{16}{n^4} = \frac{1}{\pi} \int_0^{\pi} (\pi - x)^4 dx$$

$$\frac{\pi^4}{9} + 8 \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{1}{\pi} \left\{ \frac{(\pi - x)^5}{-5} \right\}_0^{\pi} = -\frac{1}{5\pi} \{(\pi - x)^5\}_0^{\pi} = -\frac{1}{5\pi} \{0 - \pi^5\}$$

$$\Rightarrow \frac{\pi^4}{9} + 8 \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{5}$$

$$\Rightarrow 8 \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{5} - \frac{\pi^4}{9} = \frac{4\pi^4}{45}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{4\pi^4}{45 \times 8} = \frac{\pi^4}{90}$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots \dots \infty = \frac{\pi^4}{90}$$

Practice Problems

1. Find the Fourier cosine series for $f(x) = x$ in $0 < x < \pi$, show that

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}$$

$$\text{Ans: } a_0 = \pi; \quad a_n = \begin{cases} \frac{-4}{\pi n^2}, & \text{when } n \text{ is odd} \\ 0, & \text{when } n \text{ is even} \end{cases}$$

2. Find the Fourier sine series for $f(x) = x$ in $0 < x < \pi$, show that

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \dots \dots = \frac{\pi^4}{6}$$

3. Find the half range sine series for $f(x) = l - x$ in $(0, l)$. Hence prove that

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \dots \dots = \frac{\pi^4}{6}$$

HARMONIC ANALYSIS

Consider a function $y = f(x)$ with period $2l$ or 2π . Suppose $y = f(x)$ is not given explicitly. In such a case, The Fourier coefficients a_0 , a_n and b_n are not obtained directly. They can be found only approximately by numerical methods.

Assume that the function $y = f(x)$ is tabulated for equally spaced values of x in the interval $(0, 2l)$ or $(0, 2\pi)$. From this table values of x and y , we can find the constants a_0 , a_n and b_n .

FORMULAS

1. If the range $(0, 2l)$ is divided into 'n' equal sub-intervals, then the Fourier series of $f(x)$ is given by

$$y = f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

$$\text{Where, } a_0 = \frac{2}{n} \sum y, \quad a_n = \frac{2}{n} \sum y \cos \frac{n\pi x}{l} \quad \& \quad b_n = \frac{2}{n} \sum y \sin \frac{n\pi x}{l}$$

2. If the range $(0, 2\pi)$ is divided into 'n' equal sub-intervals, then the Fourier series of $f(x)$ is given by

$$y = f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$\text{Where, } a_0 = \frac{2}{n} \sum y, \quad a_n = \frac{2}{n} \sum y \cos nx \quad \& \quad b_n = \frac{2}{n} \sum y \sin nx$$

$$\text{Note : } y = f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$= \frac{a_0}{2} + (a_1 \cos x + b_1 \sin x) + (a_2 \cos 2x + b_2 \sin 2x) + \dots$$

The term $(a_1 \cos x + b_1 \sin x)$ is called the **fundamental harmonic or first harmonic**, the term $(a_2 \cos 2x + b_2 \sin 2x)$ is called the **second harmonic** and so on.

Harmonic analysis

The process of finding the Fourier series for a function $y = f(x)$ from the tabulated values of x and y at equal intervals of x is called harmonic analysis.

ILLUSTRATIVE EXAMPLES UNDER HARMONIC ANALYSIS

Example: 1 Compute the first two harmonic of the Fourier series of $f(x)$ given by the following table.

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
y	1.0	1.4	1.9	1.7	1.5	1.2	1.0

Find also the constant term.

Solution:

Here, the length of the interval is 2π

Since the last value of y is repetition of the first, only the first six values will be used, here $n = 6$

We know that the Fourier series of $f(x)$ in the interval $(0, 2\pi)$ is

$$y = f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$= \frac{a_0}{2} + a_1 \cos x + a_2 \cos 2x + b_1 \sin x + b_2 \sin 2x \quad \text{--- (1)}$$

x	y	$y \cos x$	$y \cos 2x$	$y \sin x$	$y \sin 2x$
0	1.0	1.0	1.0	0	0
$\pi/3$	1.4	0.7	-0.7	1.212	1.212
$2\pi/3$	1.9	-0.95	-0.95	1.645	-1.645
π	1.7	-1.7	1.7	0	0
$4\pi/3$	1.5	-0.75	-0.75	-1.299	1.299
$5\pi/3$	1.2	0.6	-0.6	-1.039	-1.039
Total	$\sum y = 8.7$	$\sum y \cos x = -1.1$	$\sum y \cos 2x = -0.3$	$\sum y \sin x = 0.519$	$\sum y \sin 2x = -0.173$

To find $a_0, a_1, a_2, b_1 \& b_2$ first. We know that

$$a_0 = \frac{2}{n} \sum y = \frac{2}{6} (8.7) = 2.9 \because n = 6$$

$$a_1 = \frac{2}{n} \sum y \cos x = \frac{2}{6} (-1.1) = -0.367$$

$$a_2 = \frac{2}{n} \sum y \cos 2x = \frac{2}{6} (-0.3) = -0.10$$

$$b_1 = \frac{2}{n} \sum y \sin x = \frac{2}{6} (0.519) = 0.173$$

$$b_2 = \frac{2}{n} \sum y \sin 2x = \frac{2}{6} (-0.173) = -0.058$$

Hence (1) becomes,

$$y = f(x) = \frac{2.9}{2} - 0.367 \cos x - 0.1 \cos 2x + 0.173 \sin x - 0.058 \sin 2x$$

$$= 1.45 + (-0.367 \cos x + 0.173 \sin x) + (-0.1 \cos 2x - 0.058 \sin 2x)$$

is the required Fourier series up to two harmonic.

Also constant term $a_0 = 2.9$

Example: 2 Find the Fourier series up to two harmonic for $y = f(x)$ in $(0, 2\pi)$ in the following data.

x	0°	60°	120°	180°	240°	300°	360°
y	40.0	31.0	-13.7	20.0	3.7	-21.0	40.0

Solution:

Here, the length of the interval is 360° or 2π , also $n = 6$

Since the last value of y is repetition of the first, only the first six values will be used.

We know that the Fourier series of $f(x)$ in the interval $(0, 2\pi)$ is

$$y = f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$= \frac{a_0}{2} + a_1 \cos x + a_2 \cos 2x + b_1 \sin x + b_2 \sin 2x \quad \text{--- (1)}$$

To compute a_0, a_1, a_2, b_1 & b_2 first, let us form the following tables values of x and y

x	y	$y \cos x$	$y \cos 2x$	$y \sin x$	$y \sin 2x$
0°	40.0	40.000	40.000	0	0
60°	31.0	15.500	-15.500	26.847	26.847
120°	-13.7	6.850	6.850	-11.865	11.865
180°	20.0	-20.000	20.000	0	0
240°	3.7	-1.850	-1.850	-3.204	3.204
300°	-21.0	-10.500	10.500	18.187	18.187
Total	$\sum y = 60$	$\sum y \cos x = 30$	$\sum y \cos 2x = 60$	$\sum y \sin x = 29.965$	$\sum y \sin 2x = 60.103$

We know that

$$a_0 = \frac{2}{n} \sum y = \frac{2}{6} (60) = 20 \because n = 6$$

$$a_1 = \frac{2}{n} \sum y \cos x = \frac{2}{6} (30) = 10$$

$$a_2 = \frac{2}{n} \sum y \cos 2x = \frac{2}{6} (60) = 20$$

$$b_1 = \frac{2}{n} \sum y \sin x = \frac{2}{6} (29.965) = 9.988$$

$$b_2 = \frac{2}{n} \sum y \sin 2x = \frac{2}{6} (60.103) = 20.034$$

Hence (1) becomes

$$y = f(x) = \frac{20}{2} + 10 \cos x + 20 \cos 2x + 9.988 \sin x + 20.034 \sin 2x$$

$$= 10 + (10 \cos x + 9.988 \sin x) + (20 \cos 2x + 20.034 \sin 2x)$$

Example: 3 Compute the first two harmonic of the Fourier series of $f(x)$ given by the following table.

x	0	1	2	3	4	5
y	9	18	24	28	26	20

Solution: Here, the length of the interval is $2l = 6 \Rightarrow l = 3$, Also $n = 6$

We know that the Fourier series of $f(x)$ in the interval $(0, 2l)$ is

$$y = f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$$

Put $l = 3$ in the above, we get

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{3} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{3}$$

$$= \frac{a_0}{2} + a_1 \cos \left(\frac{\pi x}{3} \right) + a_2 \cos \left(\frac{2\pi x}{3} \right) + b_1 \sin \left(\frac{\pi x}{3} \right) + b_2 \sin \left(\frac{2\pi x}{3} \right) \quad \dots (1)$$

To compute a_0, a_1, a_2, b_1 & b_2 first, let us form the following tables of values.

x	y	$y \cos(\pi x/3)$	$y \cos(2\pi x/3)$	$y \sin(\pi x/3)$	$y \sin(2\pi x/3)$
0	9	9	9	0	0
1	18	9	-9	15.588	15.588
2	24	-12	-12	20.784	-20.784
3	28	28	28	0	0

4	26	-13	-13	-22.516	-22.516
5	20	10	-10	-17.32	-17.32
Total	$\sum y$ = 125	$\sum y \cos x$ = -25	$\sum y \cos 2x$ = -7	$\sum y \sin x$ = -3.464	$\sum y \sin 2x$ = 0

We know that

$$a_0 = \frac{2}{n} \sum y = \frac{2}{6} (125) = 41.667 \therefore n = 6$$

$$a_1 = \frac{2}{n} \sum y \cos x = \frac{2}{6} (-25) = -8.333$$

$$a_2 = \frac{2}{n} \sum y \cos 2x = \frac{2}{6} (-7) = -2.333$$

$$b_1 = \frac{2}{n} \sum y \sin x = \frac{2}{6} (-3.464) = -1.155$$

$$b_2 = \frac{2}{n} \sum y \sin 2x = \frac{2}{6} (0) = 0$$

Hence (1) becomes

$$y = f(x) = \frac{41.667}{2} + (-8.333 \cos x - 1.155 \sin x) + (-2.338 \cos 2x + 0)$$

is the required Fourier series up to two harmonic.

Example: 4 Find the Fourier series up to two harmonic for the following data.

x	0	$T/6$	$T/3$	$T/2$	$2T/3$	$5T/6$	T
$f(x)$	1.98	1.30	1.05	1.30	-0.88	-0.25	1.98

Solution: Here, the length of the interval is $2l = T \Rightarrow l = T/2$, Also $n = 6$. Since the last value of y is repetition of the first, only the first six values will be used.

We know that the Fourier series of $f(x)$ in the interval $(0, 2l)$ is

$$y = f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$$

Put $l = T/2$ in the above, we get

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{\frac{T}{2}} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{\frac{T}{2}} \\ &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{2n\pi x}{T} + \sum_{n=1}^{\infty} b_n \sin \frac{2n\pi x}{T} \\ &= \frac{a_0}{2} + a_1 \cos \left(\frac{2\pi x}{T} \right) + a_2 \cos \left(\frac{4\pi x}{T} \right) + b_1 \sin \left(\frac{2\pi x}{T} \right) + b_2 \sin \left(\frac{4\pi x}{T} \right) \\ &= \frac{a_0}{2} + a_1 \cos \theta + a_2 \cos 2\theta + b_1 \sin \theta + b_2 \sin 2\theta \quad \text{--- (1)} \\ &\quad \text{Here } \theta = \frac{2\pi x}{T} \end{aligned}$$

To compute a_0, a_1, a_2, b_1 & b_2 first, let us form the following tables of values.

x	$\theta = \frac{2\pi x}{T}$	$y = f(x)$	$y \cos \theta$	$y \cos 2\theta$	$y \sin \theta$	$y \sin 2\theta$
0	0	1.98	1.980	1.980	0	0
$T/6$	$\pi/3$	1.30	0.650	-0.650	1.126	1.126
$T/3$	$2\pi/3$	1.05	-0.525	-0.525	0.909	0.909
$T/2$	π	1.30	-1.3	1.3	0	0
$2T/3$	$4\pi/3$	-0.88	0.44	0.44	0.762	0.762
$5T/6$	$5\pi/3$	-0.25	-0.125	0.125	0.217	0.217

Total		$\sum y$ = 4.6	$\sum y \cos x$ = 1.12	$\sum y \cos 2x$ = 2.670	$\sum y \sin x$ = 3.014	$\sum y \sin 2x$ = -0.328
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We know that

$$a_0 = \frac{2}{n} \sum y = \frac{2}{6} (4.6) = 1.5 \quad \{\because n = 6\}$$

$$a_1 = \frac{2}{n} \sum y \cos \theta = \frac{2}{6} (1.12) = 0.37$$

$$a_2 = \frac{2}{n} \sum y \cos 2\theta = \frac{2}{6} (2.67) = 0.890$$

$$b_1 = \frac{2}{n} \sum y \sin \theta = \frac{2}{6} (3.014) = 1.005$$

$$b_2 = \frac{2}{n} \sum y \sin 2\theta = \frac{2}{6} (-0.328) = -0.109$$

Hence (1) becomes

$$y = f(x) = 0.75 + (0.37 \cos x + 1.005 \sin x) + (0.890 \cos 2x - 0.109 \sin 2x)$$

is the required Fourier series up to two harmonic.

Practice Problems

1. Find the Fourier series up to second harmonic representing the function given by the following table.

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
y	0.8	0.6	0.4	0.7	0.9	1.1	0.8

$$\text{Ans : } a_0 = 1.5; \quad a_1 = 0.1; \quad a_2 = 0; \quad b_1 = -0.29; \quad b_2 = 0$$

$$y = f(x) = 0.75 + (0.1) \cos x - 0.29 \sin x + \dots$$

2. Compute the first two harmonic of the Fourier series for $f(x)$

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
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y	1.80	0.30	0.50	2.16	1.30	1.76	1.80
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Ans : $a_0 = 2.607$; $a_1 = -0.077$; $a_2 = 0.677$; $b_1 = -0.653$; $b_2 = -0.19$

$$y = f(x) = 1.304 - 0.077 \cos x - 0.653 \sin x + 0.677 \cos 2x - 0.19 \sin 2x + \dots$$

3. Expand the function $y = f(x)$ in the form of a Fourier series up to second harmonic from the following values of x and y.

x	0	2	4	6	8	10	12
y	9.0	18.2	24.4	27.8	27.5	22.0	9.0

Ans. $a_0 = 42.96$; $a_1 = -8.22$; $a_2 = -3.08$; $b_1 = -1.99$; $b_2 = -0.203$

COMPLEX FORM OF FOURIER SERIES

1. The complex form of Fourier series of $f(x)$ in $(-l, l)$ is

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{i n \pi x}{l}} \text{ where } c_n = \frac{1}{2l} \int_{-l}^l f(x) e^{-\frac{i n \pi x}{l}} dx$$

2. The complex form of Fourier series of $f(x)$ in $(0, 2l)$ is

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{i n \pi x}{l}} \text{ where } c_n = \frac{1}{2l} \int_0^{2l} f(x) e^{-\frac{i n \pi x}{l}} dx$$

3. The complex form of Fourier series of $f(x)$ in $(-\pi, \pi)$ is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i n x} \text{ where } c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-i n x} dx$$

4. The complex form of Fourier series of $f(x)$ in $(0, 2\pi)$ is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i n x} \text{ where } c_n = \frac{1}{2\pi} \int_0^{2\pi} f(x) e^{-i n x} dx$$

ILLUSTRATIVE EXAMPLES

Example: 1 Find the complex form of the Fourier series of e^{ax} , where 'a' is a constant in $-l < x < l$.

Solution: Let $f(x) = e^{ax}$, $-l < x < l$

The complex form of Fourier series of $f(x)$ in the interval $-l < x < l$ is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{i n \pi x}{l}} \quad \text{--- (1)}$$

$$\text{where, } c_n = \frac{1}{2l} \int_{-l}^l f(x) e^{-\frac{i n \pi x}{l}} dx$$

To find c_n :

$$c_n = \frac{1}{2l} \int_{-l}^l e^{ax} e^{-\frac{i n \pi x}{l}} dx$$

$$= \frac{1}{2l} \int_{-l}^l e^{(a - \frac{i n \pi}{l})x} dx$$

$$= \frac{1}{2l} \int_{-l}^l e^{\left(\frac{al - i n \pi}{l}\right)x} dx$$

$$= \frac{1}{2l} \left\{ \frac{e^{\left(\frac{al - i n \pi}{l}\right)x}}{\left(\frac{al - i n \pi}{l}\right)} \right\}_{-l}^l$$

$$= \frac{l}{2l(al - i n \pi)} [e^{(al - i n \pi)} - e^{-(al - i n \pi)}]$$

$$= \frac{1}{2(al - i n \pi)} [e^{al} e^{-i n \pi} - e^{-al} e^{i n \pi}]$$

$$= \frac{(-1)^n}{(al - i n \pi)} \left[\frac{e^{al} - e^{-al}}{2} \right]$$

$$\begin{aligned} \because e^{i n \pi} &= \cos n \pi + i \sin n \pi = (-1)^n + i 0 = (-1)^n \\ \because e^{-i n \pi} &= \cos n \pi - i \sin n \pi = (-1)^n - i 0 = (-1)^n \end{aligned}$$

$$= \frac{(-1)^n}{(al - i n \pi)} \sinh al \left\{ \because \sinh ax = \frac{e^{ax} - e^{-ax}}{2} \right\}$$

$$= (-1)^n \sinh a l \frac{(al + in\pi)}{(al - in\pi)(al + in\pi)}$$

$$c_n = (-1)^n \sinh a l \frac{(al - in\pi)}{a^2 l^2 + n^2 \pi^2}$$

Hence (1) becomes

$$\begin{aligned} f(x) &= \sum_{n=-\infty}^{\infty} (-1)^n \frac{(al + in\pi)}{a^2 l^2 + n^2 \pi^2} \sinh a l e^{\frac{in\pi x}{l}} \\ &= \sinh a l \sum_{n=-\infty}^{\infty} (-1)^n \frac{(al + in\pi)}{a^2 l^2 + n^2 \pi^2} e^{\frac{in\pi x}{l}} \end{aligned}$$

Example: 2 Find the complex form of the Fourier series of e^{ax} , where a is a constant in $-\pi < x < \pi$

Solution: Given $f(x) = e^{ax}$ in $-\pi < x < \pi$

The complex form of Fourier series of $f(x)$ in $(-\pi, \pi)$ is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} \quad \text{--- (1)}$$

$$\text{where, } c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

To find c_n :

$$\begin{aligned} c_n &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ax} e^{-inx} dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{(a-in)x} dx \\ &= \frac{1}{2\pi} \left\{ \frac{e^{(a-in)x}}{(a-in)} \right\}_{-\pi}^{\pi} \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2\pi} \left\{ \frac{e^{(a-in)\pi}}{(a-in)} - \frac{e^{-(a-in)\pi}}{(a-in)} \right\} \\ &= \frac{1}{2\pi} \left\{ \frac{e^{a\pi} \cdot e^{-in\pi}}{(a-in)} - \frac{e^{-a\pi} \cdot e^{in\pi}}{(a-in)} \right\} \\ &= \frac{1}{2\pi} \left\{ \frac{e^{a\pi} \cdot (\cos n\pi - i \sin n\pi)}{(a-in)} - \frac{e^{-a\pi} \cdot (\cos n\pi + i \sin n\pi)}{(a-in)} \right\} \\ &= \frac{1}{2\pi} \left\{ \frac{e^{a\pi} \cdot (-1)^n}{(a-in)} - \frac{e^{-a\pi} \cdot (-1)^n}{(a-in)} \right\} \{ \because e^{-i\theta} = \cos \theta - i \sin \theta; \cos n\pi = (-1)^n \} \\ &= \frac{(-1)^n}{\pi(a-in)} \left\{ \frac{e^{a\pi} - e^{-a\pi}}{2} \right\} \{ \because e^{-in\pi} = \cos n\pi - i \sin n\pi = (-1)^n - i0 = (-1)^n \} \\ &= \frac{(-1)^n}{\pi(a-in)} \sinh a\pi \quad \left\{ \because \sinh x = \frac{e^x - e^{-x}}{2} \right\} \end{aligned}$$

$$= \frac{(-1)^n (a + in)}{\pi(a-in)(a+in)} \sinh a\pi$$

$$c_n = \frac{(-1)^n (a + in)}{\pi(a^2 + n^2)} \sinh a\pi$$

Hence (1) becomes

$$\begin{aligned} f(x) &= \sum_{n=-\infty}^{\infty} \frac{(-1)^n (a + in)}{\pi(a^2 + n^2)} \sinh a\pi e^{inx} \\ &= \frac{\sinh a\pi}{\pi} \sum_{n=-\infty}^{\infty} \left(\frac{(-1)^n (a + in)}{(a^2 + n^2)} \right) e^{inx} \end{aligned}$$

Example: 3 Find the complex form of the Fourier series of $f(x) = \cos ax$ in

$-\pi < x < \pi$, where 'a' is not an integer.

Solution: Given $f(x) = \cos ax$ in $-\pi < x < \pi$

The complex form of Fourier series of $f(x)$ in $(-\pi, \pi)$ is given by

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} \quad \text{--- (1)}$$

$$\text{where, } c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

To find c_n :

$$\begin{aligned} c_n &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-inx} \cos ax dx \\ &= \frac{1}{2\pi} \left\{ \frac{e^{-inx}}{(in)^2 + a^2} [-incosax + asinax] \right\}_{-\pi}^{\pi} \\ &\quad \left\{ \because \int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx] \right\} \end{aligned}$$

Here $a = -in$ & $b = a$

$$\begin{aligned} &= \frac{1}{2\pi} \left\{ \frac{e^{-in\pi}}{a^2 - n^2} [-incos a\pi + asin a\pi] - \frac{e^{in\pi}}{a^2 - n^2} [-incos(-a\pi) + asin(-a\pi)] \right\} \\ &= \frac{1}{2\pi} \left\{ \frac{(-1)^n}{a^2 - n^2} [-incos a\pi + asin a\pi] - \frac{(-1)^n}{a^2 - n^2} [-incos(-a\pi) + asin(-a\pi)] \right\} \\ &\quad \{ \because e^{-in\pi} = \cos n\pi - i \sin n\pi = (-1)^n + 0 \} \\ &= \frac{1}{2\pi} \left\{ \frac{(-1)^n}{a^2 - n^2} [-incos a\pi + asin a\pi] - \frac{(-1)^n}{a^2 - n^2} [-incos a\pi - asin a\pi] \right\} \\ &= \frac{1}{2\pi} \left\{ \frac{(-1)^n}{a^2 - n^2} [-incos a\pi + asin a\pi + incos a\pi + asin a\pi] \right\} \\ &\quad \{ \because \cos(-\theta) = \cos \theta \text{ \& } \sin(-\theta) = -\sin \theta \} \\ &= \frac{1}{2\pi} \frac{(-1)^n 2 asin a\pi}{a^2 - n^2} \\ c_n &= \frac{asin a\pi}{\pi} \frac{(-1)^n}{a^2 - n^2} \end{aligned}$$

Hence (1) becomes

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{asin a\pi}{\pi} \frac{(-1)^n}{a^2 - n^2} e^{inx} = \frac{asin a\pi}{\pi} \sum_{n=-\infty}^{\infty} \frac{(-1)^n}{a^2 - n^2} e^{inx}$$

Practice problems

1. Find the complex form of the Fourier series for $f(x) = e^{-x}$; $-1 < x < 1$

$$\text{Ans: } c_n = (-1)^n \sinh 1 \frac{1 - in\pi}{1 + n^2\pi^2}$$

2. Find the complex form of the Fourier series for $f(x) = e^x$; $-\pi < x < \pi$

$$\text{Ans: } c_n = \frac{(-1)^n (1 + in)}{\pi(1 + n^2)} \sinh \pi$$

3. Find the complex form of Fourier series for $f(x) = \sin kx$, $-\pi < x < \pi$, where 'a' is not an integer.

$$\text{Ans: } c_n = \frac{(-1)^{n+1} in \sin k\pi}{\pi(k^2 - n^2)}$$

PART-A

APRIL/MAY 2017

1. State the sufficient condition for a function $f(x)$ to be expressed as a Fourier series.

2. If the Fourier series of the function $f(x) = x + x^2$, in the interval $(-\pi, \pi)$ is $\frac{\pi^2}{3} + \sum_{n=0}^{\infty} (-1)^n \left[\frac{4}{n^2} \cos nx - \frac{2}{n^2} \sin nx \right]$, then find the value of the infinite series $\frac{1}{1^2} + \frac{1}{2^2} + \dots$

PART-B

(a) (i) Find the Fourier series of period 2π for the function $f(x) = x \cos x$ in $0 < x < 2\pi$

(ii) Find the Fourier series expansion of $y = f(x)$ up to second harmonic from the following data:

x	0	1	2	3	4	5
$f(x)$	9	18	24	28	26	20

(b) (i) Find the half range cosine series of $f(x) = \begin{cases} x & \text{in } 0 < x < 1 \\ 2 - x & \text{in } 1 < x < 2 \end{cases}$

(ii) Find complex form of the Fourier series of $f(x) = e^{-ax}$, $(-l, l)$

PART-A

NOV/DEC 2016

1. State the Dirichlet's conditions for a function $f(x)$ to be expanded as a Fourier series.

2. Expand $f(x) = 1$, in $(0, \pi)$ as a Half-range sine series.

PART-B

a) (i) Expand $f(x) = x^2$ as a Fourier series in the interval $(-\pi, \pi)$ and hence deduce that $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots = \frac{\pi^4}{90}$

(ii) Obtain the constant term and the coefficient of the first sine and cosine terms in the Fourier expansion of y as given in the following table:

x	0	1	2	3	4	5
$f(x)$	9	18	24	28	26	20

(b) (i) Expand $f(x) = e^{-ax}$, $(-\pi, \pi)$ as a complex form of Fourier series

(ii) Expand $f(x) = \begin{cases} x & \text{in } 0 < x < 1 \\ 2-x & \text{in } 1 < x < 2 \end{cases}$ as a series of cosine in the interval $(0, 2)$

PART-A

MAY/JUNE 2016

3. Find the value of Fourier series $f(x) = \begin{cases} 0 & \text{in } (-c, 0) \\ 1 & \text{in } (0, c) \end{cases}$ at the point of discontinuity $x = 0$

4. Find the value of b_n in the Fourier series expansion $f(x) = \begin{cases} x + \pi & \text{in } (-\pi, 0) \\ -x + \pi & \text{in } (0, \pi) \end{cases}$

PART-B

(a) (i) Find the Fourier series $f(x) = x$ in $-\pi < x < \pi$

(ii) Find the Fourier series expansion of $f(x) = |\cos x|$ in $-\pi < x < \pi$

(b) (i) Find the half range sine series of $f(x) = x \cos \pi x$ in $(0, 1)$.

(ii) Find the Fourier cosine series up to third harmonic to represent the function given by the following data:

x	0	1	2	3	4	5
$f(x)$	4	8	15	7	6	2

PART-A

NOV/DEC 2015

1. Find the root mean square value of $f(x) = x(l-x)$ in $0 \leq x \leq l$.

2. Find the sine series function $f(x) = 1$, $0 \leq x \leq \pi$.

PART-B

(a) (i) Find the Fourier series expansion the following periodic function of period for $f(x) = \begin{cases} 2+x & -2 \leq x \leq 0 \\ 2-x & 0 < x \leq 2 \end{cases}$ Hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$.

(ii) Find the complex form of Fourier series $f(x) = e^{ax}$ in the interval $(-\pi, \pi)$ where a is a real constant. Hence deduce that $\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{a^2 + n^2} = \frac{\pi}{a \sinh \pi a}$

(b) (i) Find the half range cosine series of $f(x) = (\pi - x)^2$, $0 < x < \pi$. Hence, find the sum of series $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots$

(ii) Determine the first two harmonics of Fourier series for the following data.

x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$
$f(x)$	1.98	1.30	1.05	1.30	-0.88	-0.25

PART-A

MAY/JUNE 2015

- The instantaneous value of the function i , at time t of an alternating current wave is given by $i = I_1 \sin(\omega t + \alpha_1) + I_3 \sin(3\omega t + \alpha_3) + I_5 \sin(5\omega t + \alpha_5) + \dots$. Find the effective value of the current 'I'.
- If the Fourier series of the function $f(x) = x$, $-\pi < x < \pi$ with period 2π is given by $f(x) = 2 \left(\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \frac{\sin 4x}{4} + \dots \right)$ then find the sum of the series $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$

PART-B

(a) (i) Find the half range sine series of $f(x) = \begin{cases} x & 0 < x < \frac{\pi}{2} \\ \pi - x & \frac{\pi}{2} < x < \pi \end{cases}$. Hence deduce the sum of the series $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$.

(ii) Find the complex form of the Fourier series of $f(x) = e^{-x}$ in $-1 < x < 1$.

(b) (i) Find the Fourier series of $f(x) = |\sin x|$ in $-\pi < x < \pi$ of periodicity 2π .

(ii) Compute up to the first three harmonics of the Fourier series of $f(x)$ given by the following table.

x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π
y	1.0	1.4	1.9	1.7	1.5	1.2	1.0

PART-A

NOV/DEC 2014

- State the sufficient conditions for existence of Fourier series.

2. If $(\pi - x)^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$ in $0 < x < 2\pi$, then deduce the value of $\sum_{n=1}^{\infty} \frac{1}{n^2}$

PART-B

a) (i) Find the Fourier series of $f(x) = x^2$ in $-\pi < x < \pi$. Hence deduce the value of $\sum_{n=1}^{\infty} \frac{1}{n^2}$

(ii) Find the half range cosine series expansion of $(x-1)^2$ in $0 < x < 1$

b) (i) Compute the first two harmonics of the Fourier series of $f(x)$ from the table given.

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
$f(x)$	1.0	1.4	1.9	1.7	1.5	1.2	1.0

(ii) Obtain the Fourier cosine series expansion of $f(x) = x$ in $0 < x < 4$. Hence deduce the value of $\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots$ to ∞

PART-A

MAY/JUNE 2014

1. State the conditions for $f(x)$ to be expanded as a Fourier series in a given interval.

2. Expand $f(x) = 1$ as a half range sine series in the interval $(0, \pi)$

PART-B

(a) (i) Expand $f(x) = \begin{cases} 1 + \frac{2x}{\pi}, & -\pi < x < 0 \\ 1 - \frac{2x}{\pi}, & 0 < x < \pi \end{cases}$ as a full range Fourier series in the

interval $(-\pi, \pi)$. Hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$

(ii) Find the half range sine series of $f(x) = 4x - x^2$ in the interval $(0, 4)$. Hence

deduce the value of the series $\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \dots$

b) (i) Expand $f(x) = \sin x$ as a complex form of Fourier series in $(-\pi, \pi)$

(ii) Compute the first three harmonics of the Fourier series of $f(x)$ from the table.

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
$f(x)$	1.0	1.4	1.9	1.7	1.5	1.2	1.0

PART-A

NOV/DEC 2013

1. Find the value of a_0 in the Fourier series expansion of $f(x) = e^x$ in $(0, 2\pi)$

2. Find the half range sine series expansion of $f(x) = 1$ in $(0, 2)$

PART-B

a) (i) Find the Fourier series expansion of $f(x) = \begin{cases} 1, & 0 < x < \pi \\ 2, & \pi < x < 2\pi \end{cases}$

(ii) Find the Fourier series expansion of $f(x) = \begin{cases} -x+1, & -\pi < x < 0 \\ x+1, & 0 < x < \pi \end{cases}$

b) (i) Find the half range sine series of $f(x) = lx - x^2$ in $(0, l)$

(ii) Find the first two harmonics of the Fourier series expansion for the data

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
$f(x)$	1.0	1.4	1.9	1.7	1.5	1.2	1.0

PART-A

MAY/JUN 2013

1. State the Dirichlet's conditions for the Fourier series.

2. What is meant by Harmonic Analysis?

PART-B

b) (i) Find the Fourier series of x^2 in $(-\pi, \pi)$ and deduce that

$$\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots = \frac{\pi^4}{90}$$

(ii) Obtain the Fourier cosine series of $f(x) = \begin{cases} kx, & 0 < x < l/2 \\ k(l-x), & l/2 < x < l \end{cases}$

c) (i) Find the complex form of Fourier series of $\cos ax$ in $(-\pi, \pi)$, where ' a ' is not an integer.

(ii) Obtain the Fourier cosine series of $(x-1)^2$ in $0 < x < 1$ and hence show that $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$.